



SHEET PILING

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1.0 PURPOSE

This document provides general guidelines and recommended procedures for the analysis and design of sheet pile retaining structures.

This document reviews the theories of active and passive earth pressures as they pertain to structural design in granular and cohesive soils. These theories are then expanded to illustrate the design forces that act upon both cantilevered and anchored sheet piling systems. The unique requirements of anchored structures are examined with an emphasis placed upon tie-back design and the transmission of loads back into the supporting soil.

This document does not cover all features of sheet pile construction, but the principles presented herein may be extended to permit the engineering of more complex entities such as braced cofferdams and relieving platforms.

2.0 APPLICATION

Sheet piling is used for numerous purposes, and its versatility is invaluable in construction operations. As the name implies, piles of this type are driven closely together so that they form a continuous wall or sheet. In this arrangement, they are the backbone of various earth and water retaining structures such as breakwaters, levees, dry docks, cofferdams, bulkheads, and docks. Many uses have been made of sheet piling, and new designs will always be worked out to suit some particular situation.

3.0 INTERFACING DISCIPLINES

Not Applicable

4.0 DEFINITIONS

Not Applicable

5.0 REQUIREMENTS

Not Applicable

6.0 EXECUTION OF GUIDELINES

6.1 Design Conditions

The evaluation of the lateral forces that act on a sheet pile can be quite complex. The magnitude of earth pressure, for example, depends on the physical properties of the soil, the interaction of the soil-structure interface, the groundwater head, and the magnitude and character of the deformations in the soil-structure system. The earth pressure is also influenced by the time dependent nature of the soil strength, which varies due to creep effects and property changes within the soil.

Earth pressure against a flexible sheet pile structure is not a unique function for each soil, but rather a function of the soil-structure system. Movements of the structure are a significant factor in the distribution of earth pressure on sheet piling (refer to Attachment 01, Figure 1). The actual pressures cannot be calculated by standard theories such as those of Rankine or Coulomb; so



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empirical rules have been derived for the design of these structures. Nevertheless, the standard earth pressure theories do offer valuable expressions for the active and passive pressures in a soil mass at the state of failure. These 2 stages of stress in a soil are of particular interest in the design of sheet pile structures. When a vertical plane such as a flexible retaining wall deflects under the action of lateral earth pressure, each element of soil adjacent to the wall expands laterally, mobilizing shear resistance in the soil and causing a corresponding reduction in the lateral earth pressure.

The lowest state of lateral pressure, which is produced when the full strength of the soil is activated (a state of shear failure exists), is called the active state. The active state accompanies outward movement of the wall. On the other hand, if a vertical plane moves toward the soil such as the lower embedded portion of a sheet pile wall lateral pressure will increase as the shearing resistance of the soil is mobilized. When the full strength of the soil is mobilized, the passive state of stress exists. Passive stress tends to resist wall movements and failure.

6.1.1 Rankine Theory

The Rankine Theory is based on the assumption that the wall introduces no changes in the shearing stresses at the surface of contact between the wall and the soil. It is also assumed that the ground surface is a straight line (horizontal or sloping) and that a plane failure surface develops.

When the Rankine state of failure has been reached, active and passive failure zones will develop, as shown in Attachment 01, Figure 2.

The active and passive earth pressures for these states are expressed by the following equations:

$$P_a = rZK_a - 2c\sqrt{K_a}$$

$$P_p = rZK_p + 2c\sqrt{K_p}$$

Where

P_a and P_p = Unit active and passive earth pressures, respectively, at a depth Z below the ground surface.

rZ = Vertical pressure at a depth Z due to the unit weight, r , of soil above, using submerged weight for the soil below ground water level (i.e. vertical effective stress).

c = Soil cohesion

K_a and K_p = Coefficients of active and passive earth pressure, respectively.

Note: The letters r and B used in this document are equivalent to the Greek letters gamma and beta used in the attachments to this document.

The coefficients K_a and K_p , according to the Rankine Theory, are functions of the ϕ angle of the soil and the slope of the backfill β . For granular soils, they are given by the expressions:

$$K_a = \cos \beta \frac{\cos \beta - \sqrt{\cos^2 \beta - \cos^2 \phi}}{\cos \beta + \sqrt{\cos^2 \beta - \cos^2 \phi}}$$



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$$K_p = \cos \beta \frac{\cos \beta + \sqrt{\cos^2 \beta - \cos^2 \phi}}{\cos \beta - \sqrt{\cos^2 \beta - \cos^2 \phi}}$$

Where

ϕ = Angle of internal friction of the soil.

Note that, for the case of a level backfill, these equations reduce to:

$$K_a = \frac{1 - \sin \phi}{1 + \sin \phi} = \tan^2(45 - \phi/2).$$

$$K_p = \frac{1 + \sin \phi}{1 - \sin \phi} = \tan^2(45 + \phi/2).$$

The triangular pressure distributions for a level backfill are shown in Attachment 01, Figure 3.

It is obvious from the role played by internal shearing forces in the backfill that a cohesion component of soil shearing strength tends to decrease the active pressure and to increase the passive resistance pressure on a wall, compared with pressures computed for a soil having the same angle ϕ and zero cohesion. This conclusion is confirmed by the commonly observed fact that vertical banks of cohesive soil will in many instances stand unsupported for a considerable height and for a considerable length of time, a condition that would be impossible if the soil were cohesionless.

From the standpoint of economy in design it would appear to be logical to take cohesion into account in the determination of the overturning and translating forces that a sheet pile wall must resist. However, most designers are reluctant to do so because of the difficulty and uncertainty of determining the cohesion of the disturbed and manipulated soil that usually constitutes the backfill behind a wall. Also, the cohesion property of a disturbed soil is somewhat tenuous and may not be dependable under all climatic conditions and over a long period of time.

6.1.2 Coulomb Theory

The Coulomb Theory considers the changes resulting from frictional forces at the wall-soil interface. The roughness of the sheet piling wall commonly reduces the active and increases the passive pressure. Therefore, the assumption of a smooth wall surface (Rankine Theory) is conservative. Given the uncertainties of the actual earth pressures, the Rankine Theory is recommended for use in the design of sheet pile structures.

6.1.3 Surcharge Loads

The function of a sheet pile structure is often to retain various surface loadings as well as the soil behind it. These surface loads, or surcharge loads, also exert lateral pressures on the wall that contribute to the active pressure tending to move the wall outward. Typical surcharge loadings are railroads, highways, buildings, piles, and cranes.

The loading cases of particular interest in the determination of lateral soil pressures are the following:

1. Uniform Surcharge
2. Point Loads
3. Line Loads Parallel to the wall
4. Strip Loads Parallel to the wall



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For the case of a uniform surcharge loading, the conventional theories of earth and hydrostatic pressure can be effectively utilized. On the other hand, for point, line, and strip loads, the theory of elasticity (Boussinesq analysis modified by experiment) provides the most accurate solutions. These solutions are summarized in Teng and Terzaghi. The lateral pressures computed by these methods are conservative for sheet pile walls because they were developed based upon elastic theory and experiments on rigid, unyielding walls.

6.1.4 Uniform Surcharge

When a uniformly distributed surcharge (Q) or a live load is applied at the surface, the horizontal active pressure is increased by Qk_a .

6.1.5 Point Loads

The lateral pressure distribution on a vertical line closest to a point load may be calculated as shown in Attachment 01, Figure 4. Away from the line closest to the point load the lateral stress decreases as shown in the plan view of Attachment 01, Figure 5. Attachment 01, Figure 6 gives the lateral pressure distribution and location of the resultant force for various values of the parameter m .

6.1.6 Line Loads

A continuous wall footing of narrow width or similar load parallel to a retaining structure may be taken as a line load. For this case the lateral pressure increases from zero at the ground surface to a maximum value at some depth and gradually diminishes at greater depths. The lateral pressure distribution on a vertical plane parallel to a line load may be calculated as shown in Attachment 01, Figure 7. Attachment 01, Figure 6 gives the lateral pressure distribution and location of the resultant force for various values of the parameter m .

6.1.7 Strip Loads

Strip loads can be handled with the loading diagram and load equations indicated in Attachment 01, Figure 8.

6.1.8 Other Lateral Loads

In addition to the lateral pressures described previously, sheet pile structures may be subjected to some of the lateral loads described below.

Ice Thrust: Lateral thrusts can be caused by the volume expansion of ice in fine-grained soils (very fine sand, silt, and clay). The possibility of lateral thrust from ice or frozen ground should be eliminated by placing free-draining coarse granular soil above the frost line behind a sheet pile wall. Steel sheet piling also offers the advantage that it can yield laterally to relieve any thrust load due to ice.

Wave Forces: There are many theories concerning wave pressure against a vertical surface. In general, wave pressure is a function of wave height, length, velocity, and any other factors. The reader is directed to the works of Chellis, Anderson, and the Corps of Engineers for a detailed explanation of methods of analyses. (Refer to the REFERENCES section.)



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Ship Impact: Sheet pile dock and waterfront structures may often be subjected to the direct impact of a moving ship. Fender systems should be used in this case to spread out the reaction and reduce the impact to a minimum. Allowance for the effect of a ship's impact is sometimes made by the inclusion of an arbitrary horizontal force such as 50 to 100 tons. The reader is directed to Teng and Terghazi for further discussion.

Mooring Pull: Sheet pile dock and waterfront structures generally provide mooring posts for anchoring and docking ships. The magnitude of the mooring pull in the direction of the ship may be taken as the winch capacity used on the ship. When the spacing of the mooring posts is known, an evaluation of moor post pull on the structure can be made.

Earthquake Forces: It is generally accepted that the lateral pressures on a retaining structure increase during an earthquake, but allowances are seldom made for these dynamic loads. For critical facilities, however, where seismic effects must be considered, these loads can be evaluated by the well-known Mononobe-Okabe formulation (Seed and Whitman).

6.2 Cantilevered Sheet Piling Walls

This type of wall is suitable for moderate heights. Walls designed as cantilevers usually undergo large lateral deflections, and marine structures are readily affected by scour and erosion in front of the wall. Because the lateral support for a cantilevered wall comes from passive pressure exerted on the embedded portion, penetration depths can be quite high, resulting in excessive stresses and severe yield. Therefore, cantilevered walls using steel sheet piling are restricted to a maximum height of approximately 15 feet.

A lateral force against a cantilevered wall is illustrated in Attachment 01, Figure 9. When the load (P) is applied to the top of the wall, the piling rotates about the pivot point b , mobilizing passive pressure above and below the pivot point. The term ($P_p - P_a$) is the net available passive pressure since both are acting on the wall. If applicable, the difference in water table on each side of the sheet piling should also be considered.

At point b , the piling does not move and would be subjected to equal and opposite at-rest earth pressures with a net pressure equal to zero. The resulting earth pressure is represented by the diagram $oabc$. For the purpose of design, the curve abc is replaced by a straight line dc . The point d is located so as to make the sheet piling in a state of static equilibrium. Although the assumed pressure distribution is in error, it is sufficient for design purposes.

The distribution of earth pressure is different for sheet piling in granular soils and sheet piling in cohesive soils. Also, the pressure distribution in clays is likely to change with time. Therefore, the design procedures for sheet piling in both types of soils are discussed separately.

6.2.1 Cantilever Sheet Piling in Granular Soils

The following table may be used to estimate a trial depth of penetration, D .

Standard Penetration Resistance, N Blows/Foot	Relative Density of Soil	Depth of Penetration*
0-4	Very Loose	2.00 H
5-10	Loose	1.50 H
11-30	Medium Dense	1.25 H
31-50	Dense	1.00 H
+50	Very Dense	0.75 H



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*H = Height of piling above the dredge line

The active and passive lateral pressures can now be determined using the appropriate coefficients. The resulting earth pressure diagram for a homogeneous granular soil is shown in Attachment 01, Figure 10 where the active and passive pressures are overlain to describe pictorially the resulting soil reactions. Note that the top of the ground surface is horizontal.

The requirement of static equilibrium must be satisfied; the sum of the forces in the horizontal direction must be zero, and the summation of moments about any point must be zero. The sum of the horizontal forces may be written in terms of pressure areas:

$$(EA_1A_2) - (FBA_2) - (ECJ) = 0$$

The distance Z may be determined from the above equation. For a uniform granular soil,

$$Z = -\frac{K_p D^2 - K_a (H + D)^2}{(K_p - K_a)(H + 2D)}$$

Moments should be taken about the point F to determine whether the summation of moments is zero, as it must be. If not, the depth of penetration, D , must be adjusted and the above steps repeated until convergence is reached; that is, the sum of the moments about F is zero.

When equilibrium is established between the active and passive soil pressures, 20 to 40 percent should be added to the calculated depth of penetration, D . This will give a safety factor of approximately 1.5 to 2.0. An alternative and more desirable method is the use of a reduced value of the passive earth pressure coefficient for design. The maximum allowable earth pressure should be limited to 50 to 75 percent of the ultimate passive resistance.

Before increasing the depth of penetration, the maximum moment in the sheet piling should be calculated at the point of zero shear.

A rough estimate of the lateral displacement may be obtained by considering the wall to be rigidly held at an embedment of $1/2 D$ and subjected to a triangular load distribution approximating the actual applied active loading. The displacement in inches at any distance y from the top of the pile is then given by the following expression:

$$\frac{P_t}{60EIL^2} (Y^5 - 5L^4 + 4L^5)$$

Where

P_t = total applied load over length in pounds

L = $H + 1/2 D$ in inches

In which

H = exposed length of sheeting in inches

And

D = Penetration of sheeting in surface stratum, plus $1/2$ of penetration in any lower, more dense, coarse grained stratum (inches). Neglect any penetration in rock.



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6.2.2 Cantilever Sheet Piling in Cohesive Soils

Two cases of cantilevered walls in cohesive soils are of interest: (1) sheet pile walls entirely in clay and (2) walls driven in clay and backfilled with sand. Different lateral earth pressures develop for each case.

- Wall Entirely in Cohesive Soil

The design of sheet piling in cohesive soils is complicated by the fact that the strength of clay changes with time and, accordingly, the lateral earth pressures also change with time. The depth of penetration and the size of piling must satisfy the pressure conditions that exist immediately after installation and the long-term conditions after the strength of the clay has changed. Immediately after the sheet piling is installed, the earth pressure may be calculated on the assumption that undrained strength of the clay prevails. That is, it is assumed that the clay derives all its strength from cohesion and no strength from internal friction. The analysis is usually carried out in terms of total stress using a cohesion value, c , equal to 1/2 the unconfined compressive strength, q_u . The method is usually referred to as a $\phi = 0$ analysis.

Attachment 01, Figure 11 illustrates the initial pressure conditions for sheet piling embedded in cohesive soil for its entire depth.

When $\phi = 0$, the ultimate passive earth pressure on the left side of the piling, per Rankine Theory, is

$$P_p = r_e (Z - H) + q_u$$

and the active pressure on the right side of the piling is given by

$$P_a = r_e Z - q_u$$

Where

Z = depth below the original ground surface in feet.

q_u = unconfined compressive strength, pounds per square foot ($q_u = 2c$).

r_e = effective unit soil weight (moist unit weight above the water level and submerged unit weight below the water level), pounds per cubic foot.

The negative earth pressure of tension zone, as shown by the dashed line, is ignored because the soil may develop tension cracks in the upper portion. Because the slopes of the active and passive pressure lines are equal ($K_a = K_p$), the net resistance on the left side of the wall is constant below the dredge line and is given by

$$P_p - P_a = 2q_u - r_e H$$

Note that theoretically there will be no net pressure, and the wall will fail if $r_e H$ is equal to or greater than $2q_u$. The height $H_c = 2q_u / r_e$ is often called the critical wall height. Design criteria for checking the stability of the wall are given in Anchored Sheet Piling in Cohesive Soils, Attachment 01, Figure 15.

For the lower portion, where the piling moves to the right, the net resistance is given by

$$P_p - P_a = 2q_u + r_e H$$

which is illustrated in Attachment 01, Figure 11 (b).



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The resulting net pressure distribution on the wall is as shown in Attachment 01, Figure 11 (a), and the method of solution is the same as that presented for the design of cantilevered sheet pile walls in granular soils. The point d and the depth of penetration D are chosen to satisfy the conditions of static equilibrium; that is, the sum of the horizontal forces is equal to zero and the sum of the moments about any point is equal to zero. Finally, D should be increased by 20 to 40 percent to obtain the total required depth of penetration.

- Wall in Cohesive Soil below the Dredge Line: Granular Backfill above the Dredge Line

The above method may be extended to the case in which sheet piling is driven in clay and back filled with granular soil as shown in Attachment 01, Figure 12. The only difference is that the active pressure above the dredge line is equal to $K_a \gamma_e Z$ for a granular material. The methods of design are exactly the same as discussed previously. The long-term condition for sheet piling in clays must also be considered due to the time dependent changes in ϕ and c . The analysis should be carried out using effective stress parameters c' and ϕ' , obtained from consolidated drained tests, or from consolidated undrained tests in which pore pressure measurements are made. Limited experimental data indicate that the long-term value of c' is quite small, and that for design purposes c' may be conservatively taken as zero. The final value of ϕ' is usually between 20 and 30 degrees. The lateral pressures in the clay over a long period of time approach those for a granular soil. Therefore, the long-term condition is analyzed as described in the preceding section for granular soils.

6.3 Anchored Sheet Piling Walls

Anchored sheet pile walls derive their support by 2 means: passive pressure on the front of the embedded portion of the wall and anchor tie rods near the top of the piling. This method is suitable for heights up to about 35 feet, depending on the soil conditions. For higher walls the use of high-strength steel piling, reinforced sheet piling, relieving platforms, or additional tiers of tie rods may be necessary. The overall stability of anchored sheet pile walls and the stresses in the members depend on the interaction of a number of factors, such as the relative stiffness of the piling, the depth of piling penetration, the relative compressibility and strength of the soil, and the amount of anchor yield. In general, the greater the depth of penetration, the lower the resultant flexural stresses.

Attachment 01, Figure 13 shows the general relationship between depth of penetration, lateral pressure distribution, and elastic line or deflection shape.

Case (a) is commonly called the free earth support method. The passive pressures in front of the wall are insufficient to prevent lateral deflection and rotations at point C. Cases (b), (c) and (d) show the effect of increasing the depth of penetration. In cases (b) and (c) the passive pressure has increased enough to prevent lateral deflection at C; however, rotation still occurs. In case (d) passive pressure has sufficiently developed on both sides of the wall to prevent both lateral deflection and rotation at C. This case is commonly called the fixed earth support method because point C is essentially fixed. Cases (a) and (d) represent the 2 extremes in design.

Some analytical techniques in current usage for the design of anchored sheet pile walls are the following:

- Free Earth Support Method
- Fixed Earth Support Method (Equivalent Beam)
- Graphical Methods
- Danish Rules



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The Free Earth Support Method is based on the assumption that the soil into which the lower end of the piling is driven is incapable of producing effective restraint from passive pressure to the extent necessary to induce negative bending moments. The piling is driven just deep enough to provide stability, assuming that the maximum possible passive resistance is fully mobilized. The sheet piling is assumed to be inflexible so that no pivot point exists below the dredge line (that is, no passive resistance develops on the backside of the piling). With these assumptions the design becomes a problem in simple statics. Procedures for the design of anchored sheet piling in granular and cohesive soils are discussed separately below.

The Fixed Earth Support Method requires the piling to be driven deep enough so that at some point below the dredge line its lateral deflection and rotation are zero. The procedure of assuming a depth of penetration and calculating the resulting deflected shape to confirm that it agrees with the assumption is very time consuming and very seldom used in practice. A simpler procedure known as the Equivalent Beam Method is valid only for sheet piling driven in granular soil and backfilled with granular soil.

Graphical Methods can sometimes be used advantageously to design sheet piling walls for cases of complex or irregular loading. The reader is referred to The U.S. Steel Sheet Piling Design Manual.

The Danish Rules are purely empirical and apply only to anchored sheet pile walls in granular material. They represent the least conservative approach to design and are not recommended for use in design.

6.3.1 Anchored Sheet Piling in Granular Soil

The active and ultimate passive lateral pressures are computed as before, using the appropriate coefficients. Note that Attachment 01, Figure 14 shows a generalized case for an anchored wall entirely in granular material but having different soil properties. Therefore, r_e refers to the equivalent soil unit weight, either wet or submerged, for the particular soil layer in question. Also, K'_a refers to the active pressure coefficient for the natural in-place soil.

The weight of overburden and surcharge load at the dredge level is equal to $r_e H$ and the point of zero pressure is given by

$$y = \frac{r_e H K'_a}{P_p - P_a}$$

The resultant force of the earth pressure above point a, P_a , and its distance L below the tie rod level may now be determined. To satisfy equilibrium, the wall must be deep enough so that the moment due to the net passive pressure will balance the moment due to the resultant active force, P_a . This requires that

$$\frac{(P_p - P_a)}{2} D_1^2 \left(H_t + Y + \frac{2D_1}{3} \right) = L P_a$$

which of course is a cubic equation in terms of D_1 . D_1 may be evaluated by trial and error.

The tie rod tension per lineal length of wall is then expressed by

$$T = P_a - \frac{(P_p - P_a)}{2} D_1^2$$



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The maximum bending moment will occur at the point of zero shear in the wall below the rod level. 20 to 40 percent should be added to D_1 to provide a margin of safety; or, alternatively, the maximum allowable passive pressure should be limited to 50 to 75 percent of the ultimate passive pressure in the initial computations.

6.3.2 Anchored Sheet Piling in Cohesive Soils

Attachment 01, Figure 15 shows the resulting pressure diagram for an anchored sheet pile wall in cohesive soil. The immediate and long term strengths of the soil are established from undrained test results ($\phi = 0$) and drained test results ($c = 0$), respectively. The design height should be checked to see whether if the stability of the wall complies with the criteria that will be presented in the discussion immediately following this section. If the stability is acceptable, the resultant force P_a , due to active earth pressure (and any surcharge) above the dredge line, may now be calculated. To satisfy equilibrium, the summation of moments about the tie rod level requires that

$$(4c - r_e H)D' \left(H_t + \frac{D'}{2} \right) = LP_a$$

which determines D' . The tie rod tension per lineal length of wall is then expressed by

$$T = P_a - (4c - r_e H)D'$$

The design moment in the sheet piling may now be calculated at the point of zero shear occurring below the tie rod level. To provide an adequate safety factor, D' should be increased by 20 to 40 percent: an acceptable alternative would be to limit the cohesion to 50 to 75 percent of the full cohesion at the onset of the computations.

- Stability of Sheet Pile Walls

The height of a sheet pile wall driven in cohesive soils is limited by the initial strength of the clay below the level of the dredge line. This is true for anchored or cantilevered walls and for either granular or cohesive backfill above the dredge line. For heights in excess of this limit, the wall will fail. Therefore, the first step in the design of sheet pile walls in cohesive soils should be the investigation of the limiting height.

Attachment 01, Figure 16 shows a sheet pile wall driven in cohesive soil together with the lateral earth pressures below the dredge line. The net passive resistance below the dredge line is given by this formula:

$$P_p - P_a = 2q_u - r_e H$$

If the height of the wall, H , is such that the net passive resistance is zero, failure will occur. This will occur when $2q_u = r_e H$, that is, when the ratio $2q_u/r_e H = 1$. The stability number, S , is defined as

$$S = \frac{q_u}{2r_e H} = \frac{c}{r_e H}$$

Because adhesion, c_a , will develop between the soil and the sheet piling, the stability number may be modified to be

$$S = \frac{c}{r_e H} \sqrt{(1 + c_a/c)}$$



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For design, it is sufficient to take the value of $(1 + c_a/c)^{1/2}$ equal to 1.25 and, therefore, $S = 0.3$. Hence, a sheet pile wall driven into cohesive soils should have a minimum stability number of about 0.3 times an appropriate safety factor.

6.3.3 Tie Rods

Tie rods are frequently subjected to tension much greater than the calculated values. The conventional methods of calculating anchor pull involve the assumption that the resulting active pressure distribution is hydrostatic, or triangular. In reality, the real distribution may be somewhat different, and the corresponding anchor tension may be greater than that computed. The anchor pull may also increase because of repeated application and removal of heavy surcharges or an unequal yield of adjacent anchorages that causes over-loading. Because of these possibilities, the computed tie rod design tension should be increased by about 30 percent for the tie rod proper, and 50 to 100 percent at splices and connections where stress concentrations can develop. The pull on a tie rod before any increase is assessed, is

$$A_p = \frac{Td}{\cos x}$$

where

- A_p = The anchor pull in pounds per tie rod
- T = The anchor pull in pounds per foot of length
- d = Distance between rods in feet (center to center)
- x = Inclination of tie rod with the horizontal

Any soft soil below the tie rods, even at great depth, may consolidate under the weight of recent backfill, causing the ground to settle. A small settlement will cause the tie rods to sag under the weight of the soil above them. This sagging will result in an increase in tensile stress in the tie rod as it tends to pull the sheeting. In order to eliminate this condition, the tie rods should be supported with light vertical piles at 20 to 30 foot intervals or be encased in large conduits.

6.3.4 Wales

The horizontal reaction from an anchored sheet pile wall is transferred to the tie rods by a flexural member known as a wale. It normally consists of 2 spaced structural steel channels placed with their webs back-to-back in the horizontal position. Attachment 01, Figure 17 shows common arrangements of wales and tie rods located on both the inside and outside of a sheet pile wall. The channels are spaced with a sufficient distance between their webs to clear the upset end of the tie rods. Pipe segments or other types of separators are used to maintain the required spacing when the channels are connected together. If wales are constructed on the inside face of the sheet piling, every section of sheet piling is bolted to the wale to transfer the reaction of the piling. Although the best location for the wales is on the outside face of the wall, where the piling will bear against the wales, they are generally placed inside the wall to provide a clear outside face. For sump pits, inside whalers and tie rods would need to provide clearance for wall construction.

For sizing purposes, the response of a wale may be assumed to be somewhere between that of a continuous beam on several supports (the tie rods) and a single span on simple supports. Therefore, the maximum bending moment for design will be somewhere between



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$$M_{\max} = (1/10)Td^2 \text{ (3 continuous spans simply supported)}$$

$$M_{\max} = (1/8)Td^2 \text{ (single span simply supported)}$$

where

T = the anchor pull in pounds per foot (before increase)

d = distance between rods in feet (center to center)

The above expressions are only approximations. An exact analysis would have to take into account the overall sheet pile layout, elasticity of the tie rods, the rigidity of the wale, and the residual stresses induced during bolting operations.

Wales are connected to the sheet piling by means of fixing plates and bolts. Each bolt transmits a pull proportional to the width, ℓ , of a single sheet pile, and equal to

$$R_b = T \times \ell \times F.S.$$

where

R_b = pull in pounds per bolt

ℓ = the driving distance of a single sheet pile (if each section is bolted)

$F.S.$ = a desired safety factor to cover stresses induced during bolting (between 1.2 and 1.5)

The fixing plate (as shown in Attachment 01, Figure 17, Section A-A) may be designed as a beam simply supported at 2 points (the longitudinal webs of the wale) and bearing a single load, R_b , in the center.

The wales are field bolted at joints known as fish plates or splices, as shown in Attachment 01, Figure 17, Section C-C. It is preferable to splice both channels at the same point and place the joint at a recess in the double piling element. Splices should be designed for the transmission of the bending moment.

6.3.5 Anchors

The stability of an anchored sheet pile bulkhead depends mainly on the anchor device to which the wall is fastened. The reaction of the tie rods may be carried by any one of the types of anchorages shown in Attachment 01, Figure 18.

In order for an anchorage system to be effective, it must be located outside the potential active failure zone developed behind a sheet pile wall. Its capacity is also impaired if it is located in unstable ground or if the active failure zone prevents the development of full passive resistance of the system. Attachment 01, Figure 19 shows several installations that will not provide the full anchorage capacity required because of failure to recognize the above considerations.

Attachment 01, Figure 20 shows the effect of anchorage location on the resistance developed.

If the anchorage is located between bc and bf , only partial resistance is developed due to the intersection of the active and passive failure wedges. However, the theoretical reduction in anchor capacity may be analytically determined as explained in Terghazi, Theoretical Soils Mechanics.



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- Sheet Pile Anchor Walls

Short steel sheet piles driven in the form of a continuous wall may be used to anchor tie rods. The tie rods are connected with a waling system similar to that for the parent wall, and resistance is derived from passive pressure developed as the tie rod pulls against the anchor wall. To provide some stability during installation of the piling and the wales, pairs of the piling should be driven to a greater depth at frequent intervals. The anchor wall is analyzed by conventional means considering full passive pressure developed only if the active and passive failure zones do not intersect. However, if the failure wedges do intersect, the total passive resistance of the anchor wall will be reduced by the amount

$$P_p = (K_p - K_a) \frac{r h_2^2}{2} \quad (\text{for granular soils})$$

where

h_2 = depth to the point of intersection of the failure wedges as shown in Attachment 01, Figure 21(b).

Ideally, the tie rod connection should be located at the point of the resultant earth pressure acting on the wall.

- Deadmen Anchors

The effects of interaction of the active and passive failure surfaces, as mentioned above, also apply to the design of deadmen anchors.

Care must be exercised to see that the anchor block or deadman does not settle after construction. This is generally not a problem in undisturbed soils; however, where the anchorage must be located in unconsolidated fill, piles may be needed to support the blocks. Also, the soil within the passive wedge of the anchorage should be compacted to at least 90 percent of maximum modified dry density unless the deadman is forced against firm natural soil.

A continuous deadman is shown in Attachment 01, Figure 22. If $H/2$ is greater than h , assume that the deadman extends to the ground surface and the ultimate capacity of the deadman is

$$T_{ult} = P_p - P_a$$

where

T_{ult} = Ultimate capacity of the deadman, pounds per linear foot

P_p = Total passive earth pressure, pounds per linear foot

P_a = Total active earth pressure, pounds per linear foot

The active and passive pressure distributions for granular and cohesive soils are also shown in Attachment 01, Figure 22. For design in cohesive soils, both the immediate and the long-term pressure conditions should be checked to determine the critical case. A safety factor of 2 against failure is recommended

$$T \leq T_{ult} / 2$$

Attachment 01, Figure 23 shows a deadman of length L located near the ground surface and subjected to an anchor pull T . Experiments have indicated that at the time of failure, due to edge effects, the heave of the ground surfaces takes place in an area as shown. The surface of sliding at both ends is curved. Integration of the resistance along these curved sliding surfaces results in the following expression for the ultimate capacity of short deadmen in granular soils.



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$$T_{ult} \leq L(P_p - P_a) + \frac{K_o r}{3} (K_p^{1/2} + K_a^{1/2}) H^3 \tan \phi$$

where

- T_{ult} = Ultimate capacity of the deadman, pounds
- L = Length of the deadman, feet
- P_p, P_a = Total passive and active pressure, pounds per lineal foot
- K_o = Coefficient of earth pressure at rest. (It may be taken as 0.4 for design of deadman)
- r = Unit weight of soil, pounds per cubic foot
- K_p, K_a = Coefficients of passive and active earth pressure
- ϕ = Angle of internal friction
- H = Height of deadman, feet

For cohesive soils, the second term in the above expression should be replaced by the cohesive resistance, thus

$$T_{ult} \leq L(P_p - P_a) + 2cH^2$$

where

- c = The cohesion of the soil, pounds per square foot

- **Pile A-Frames**

Brace piles forming A-frames can sometimes be used effectively to anchor sheet pile walls, as shown in Attachment 01, Figure 18(b). If only 2 piles form each frame, it is necessary to connect the frames with a continuous reinforced concrete cap. The anchor rods can then be attached to the concrete cap. However, if 3 piles are used, each frame can support a tie rod through the center pile and act independently. The pile angled toward the wall will be in compression whereas the pile or piles angled away from the wall will be in tension. The resulting forces are easily determined from a force polygon as shown in Attachment 01, Figure 18 (b). This method of support can be used effectively only if the brace piles can be adequately seated in a underlying stratum of soil or, preferably, rock.

- Additional types of anchoring include passive grouted anchors, post-tensioned grouted anchors, and screw anchors. Geotechnical assistance and/or contractor assistance should be requested to assist in the design.

6.4 Types of Sheet Piling

6.4.1 Timber Sheet Piling

Timber sheet piling is used for short spans, light lateral loads, and commonly, for temporary structures in the form of braced sheeting. If it is used in permanent structures above water level, it requires preservative treatment; and even then, the useful life is relatively short.

Timber piling probably finds its greatest use as braced sheeting for temporary retaining structures in excavations.



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Driving of wood sheeting is somewhat troublesome because a driving cap is required; and driving in hard soil with large gravel tends to split the piling. The lower end of the pile may be cut with a bevel and provided with a driving shoe made of 1/16 to 1/8 inch steel. The sheeting is joined, generally, as shown in Attachment 01, Figure 24, and placed so that the piling tends to wedge against the previously driven pile. The tongue-and-groove joints indicated will provide a reasonably well jointed wall only if small stones or soil do not become wedged into the grooves. When splines are used, the sheets are grooved in the mill and the splines are driven after the piles are in place.

6.4.2 Concrete Sheet Piling

Concrete sheet piles are relatively rigid precast members designed to withstand permanent stresses during service and handling stresses during construction. The procedure for determination of permanent stresses has been described in the discussion on design methods. The handling stresses are produced by the weight of the pile when it is picked up in a more or less horizontal position. A short pile may be handled with a sling looped approximately at the third point from the top. For long piles, 2 or more pickup points may be necessary in order to reduce the handling stresses. The locations of pickup points must be clearly marked on the piles. To avoid extensive cracks due to shrinkage and handling, a certain minimum amount of reinforcing is desirable, as shown in Attachment 01, Figure 25, even if the calculated stresses are low.

Also the reinforcing ties should be placed at closer spacing at the top and bottom of the pile to reduce the possibility of damage due to driving impact. The bottom of the pile is usually made with a bevel on the tongue side so that the pile will tend to be located tightly against the one previously driven. Sheet piling walls are generally provided with flexibility to allow for relative movement between sections of wall. If grouted for water tightness, concrete piles should have expansion joints above the dredge line at intervals of 25 to 75 feet. It may be convenient to cast a special unit that is solid below ground and split above, the gap being filled with flexible joint filler. A cap is generally placed on the sheet pile wall, and the flexible joint should continue up through the cap.

Concrete sheet piles are heavy and bulky, and therefore they require heavier equipment to handle and drive. Because of their large volume of displacement they also encounter greater driving resistance.

6.4.3 Steel Sheet Piling

Steel sheet piles are rolled structural members with interlockings to engage with one another. Their principal advantages over other sheet pilings are their high resistance to driving stresses, relatively light weight, potential for reuse, and their ability to be lengthened either by welding or bolting.

The geometry of pile sections is quite diverse. Attachment 01, Figures 26(a) and (b) illustrate some typical Bethlehem and European sheet pile cross sections.

The Bethlehem sections employ 2 types of interlocks: finger-and-thumb and ball-and-socket. The interlocks take somewhat different forms in sheet piles produced by different manufacturers, and often only a few sections manufactured by the same producer will interlock with each other. Therefore, if different sections are contemplated for the same job, the manufacturers' catalogues should be consulted.

In American practice, the interlocks are assumed as offering no frictional resistance, and the section modulus is calculated about the neutral axis of each pile. There is undoubtedly a certain



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amount of friction between piles, and the actual flexural strength is therefore larger than the calculated value. All steel sheet piles are normally rolled in mills. Pieces at corners and joints, in the forms of *Y* or *T* are fabricated, either by bolting, riveting, or welding. Attachment 01, Figure 27 shows some typical details.

Caps, Attachment 01, Figure 28, for steel sheet piling walls may be steel plates or rolled sections, reinforced concrete, or wood. The space between upturned flanges should be filled with concrete to prevent corrosion; wood should be treated.

6.4.4 Durability of Steel Sheet Piling

In most normal conditions applicable to sheet pile structures, three possible corrosion situations should be considered: corrosion due to soil, to water, and to the atmosphere.

When the piling is below groundwater level and is in contact with undisturbed soil, corrosion is so slight that it can often be neglected, and it is generally unnecessary to apply any form of protective coating. Above the ground water level some corrosion may occur, but the usual practice of allowing an additional thickness of 1/16 of an inch for corrosion is satisfactory in normal conditions. Only in occasional cases will it be necessary to remove corrosive soils or provide cathodic protection.

When completely immersed, the average corrosion of bare steel does not usually exceed 0.003 inches per year in sea water, and 0.002 inches per year in fresh water. These rates are likely to be exceeded in the splash zone between tide levels and above high water, but this area is usually accessible for repainting.

Corrosion of steel due to the atmosphere can be quite variable and may be high in some circumstances; but coatings may be used, and it will usually be possible to carry out maintenance.

Unusual soil, water, or atmospheric conditions can give rise to significant corrosion. Thus, for example, special consideration should be given in tropical conditions, or where aggressive chemicals are present, or where the steel is in contact with other metals.

No significant differences in corrosion rates have been observed for the various types of steel normally used for the manufacture of steel sheet piling. The inclusion of a copper content has not been found to have any noticeable effect in reducing corrosion due to water or soil, but copper contents of the order of 0.3 percent and higher can give some improvement in atmospheric corrosion resistance when the steel is uncoated.

6.4.5 Coatings

The economics of painting steel pile structures is quite different from that of painting aboveground steel structures because the parts of the piling that are below water or soil level are not accessible for future maintenance. Because of this factor, the life of the piling will be determined by the life of the initial coating, which may be only a few years; and after that the corrosion rate of the steel itself will become the governing factor. A common practice for steel piling is to apply 1 or 2 heavy coats of a coal tar epoxy paint without special surface preparation other than the removal of dirt and loose mill-scale by wire brushing. The very high costs of application of special coatings, normally involving several coats after sand-blasting, are almost never justified, because such coatings will only delay the onset of corrosion by a few extra years compared to simpler coatings.



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If the expected life of a particular sheet pile wall is 50 years, it may not be worth spending an extra 15 to 25 percent of the cost of the piling on a special coating that may only increase the life of the piling by an extra 5 years or so. In some circumstances, the designer may wish to consider the use of concrete sheet piles when a long structural life is required. In normal corrosion conditions, it is clearly cheaper to use a slightly increased thickness of steel, if indeed the extra few years of life are significant.

Where coatings are used, the selection of type should take into account their maintenance where the surfaces are accessible; for example, paints that require a perfectly clean and dry surface for their application will not be suitable for painting between tides. Paints may be damaged during driving, particularly in gravel soils, or during transport and handling of the piles, or when the piles are in contact with guide walings used for support while driving.

Steel walings, tie rods, and other steel fittings can be protected with simple coal tar or epoxy based paints, preferably 2 coats, whereas tie rods should be spirally wrapped with a durable fabric or fiberglass tape after the application of the first coat.

7.0 RESOURCES

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9.0 ATTACHMENTS, ADDENDA

Attachments

Attachment 01

Figure 1. Distribution of Lateral Earth Pressure

Figure 2. Rankine Failure Zones



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Figure 3. Summary of Commonly Used Rankine Formulae

Figure 4. Lateral Pressure Due to Point Load (Elevation)

Figure 5. Lateral Pressure Due to Point Load (Plan)

Figure 6. Lateral Pressure Distribution and Location of Resultant Force Due to Point or Line Load

Figure 7. Lateral Pressure Due to Line Load

Figure 8. Lateral Pressure Due to Strip Load

Figure 9. Earth Pressure on Cantilever Sheet Piling

Figure 10. Earth Pressure for Design of Cantilever Sheet Piling in Granular Soil

Figure 11. Initial Earth Pressure for Design of Cantilever Sheet Piling Entirely in Cohesive Soil

Figure 12. Initial Earth Pressure for Design of Cantilever Sheet Piling in Cohesive Soil with Granular Backfill

Figure 13. Effect of Depth of Penetration on Pressure Distribution and Deflected Shape

Figure 14. Earth Pressure for Design of Anchored Sheet Piling in Granular Soil by the Free Earth Support Method

Figure 15. Initial Earth Pressure for Design of Anchored Sheet Piling in Cohesive Soil by the Free Earth Support Method

Figure 16. Stability of Sheet Piling in Cohesive Soils

Figure 17. Typical Wale and Anchor Rod Details

Figure 18. Types of Anchorage Systems

Figure 19. Installations Having Reduced Anchorage Capacity

Figure 20. Effects of Anchor Location Relative to the Wall

Figure 21. Continuous Anchor Walls

Figure 22. Continuous Deadmen Near Ground Surface

Figure 23. Short Deadmen Near Ground Surface

Figure 24. Wood Sheet Piles

Figure 25. Concrete Sheet Piles

Figure 26. Steel Sheet Piles: (a) Some Bethlehem Shapes, (b) Some European Shapes

Figure 27. Typical Fabricated or Rolled Steel Sheet Pile Joints

Figure 28. Typical Caps for Steel Sheet Piling

Sample Design 1: Design of Cantilevered Sheet Pile Wall - Granular Soil

Attachment 02



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Attachment 03	Sample Design 2: Design of Cantilevered Sheet Pile Wall - Cohesive Soil
Attachment 04	Sample Design 3: Design of Anchor Sheet Pile Wall - Cohesive Soil With Sand Backfill - Free Earth Support Method Resultant Pressure Distribution
Attachment 05	Sample Design 4: Design of Anchored Sheet Pile Wall - Granular Soil Free Earth Support Method
Addenda	
None	

SHEET PILING

Figure 1. Distribution of Lateral Earth Pressure

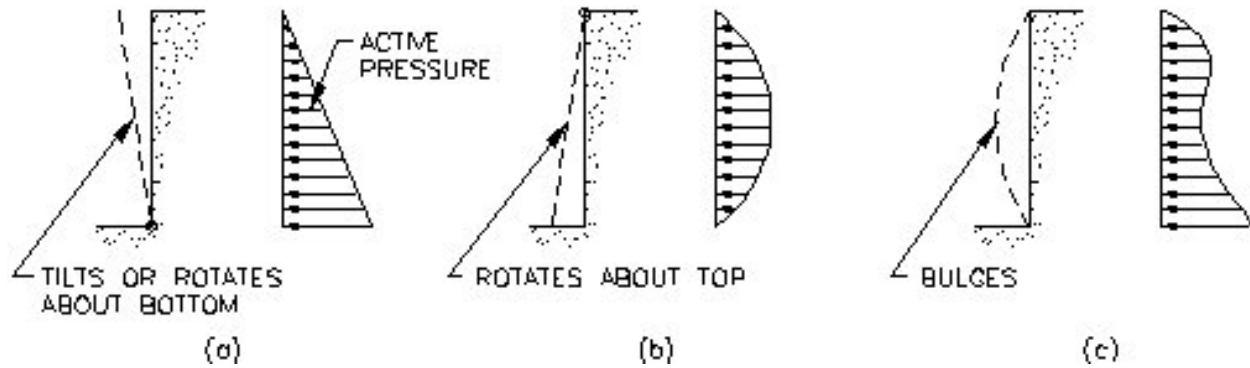
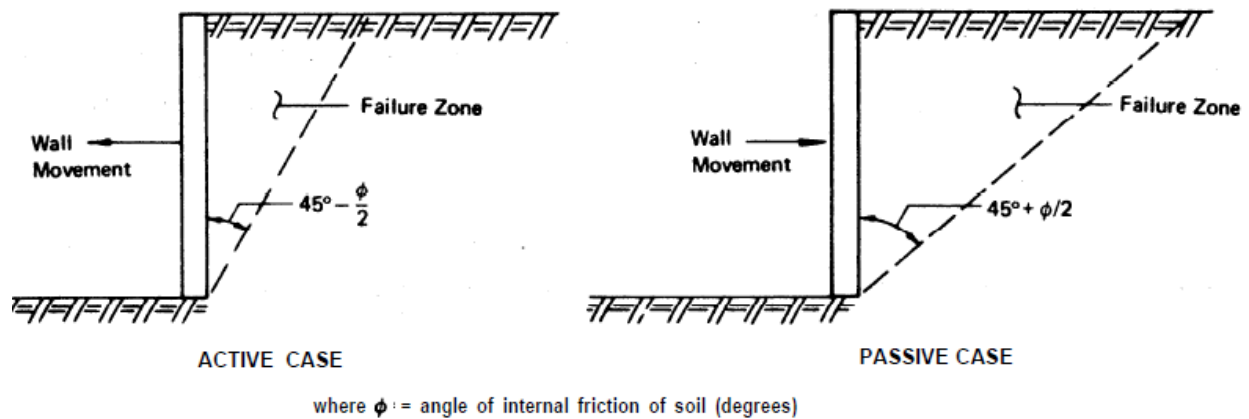


Figure 2. Rankine Failure Zones



SHEET PILING

Figure 3. Summary of Commonly Used Rankine Formulas

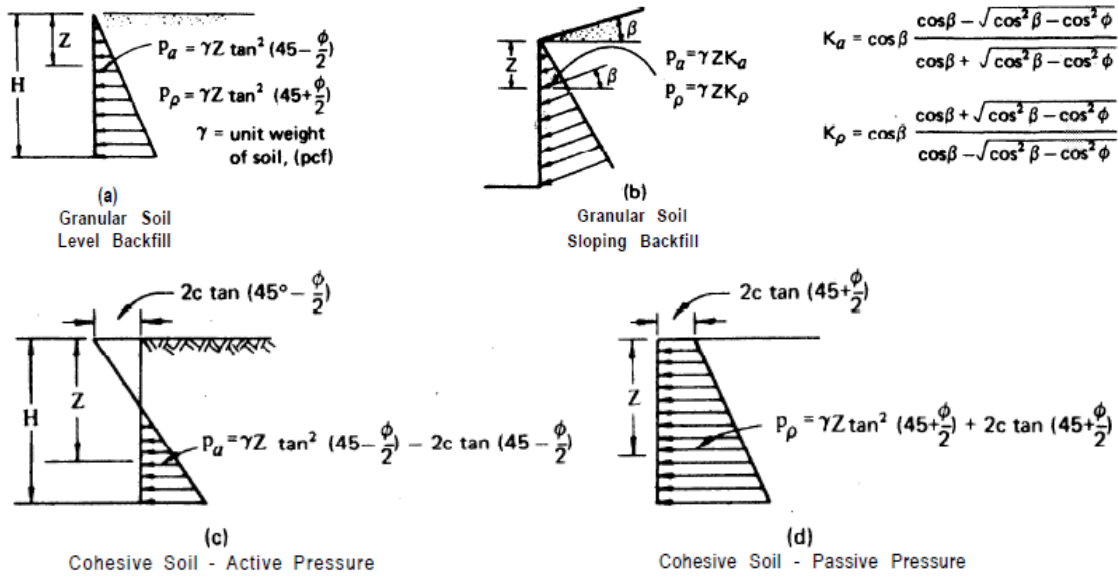
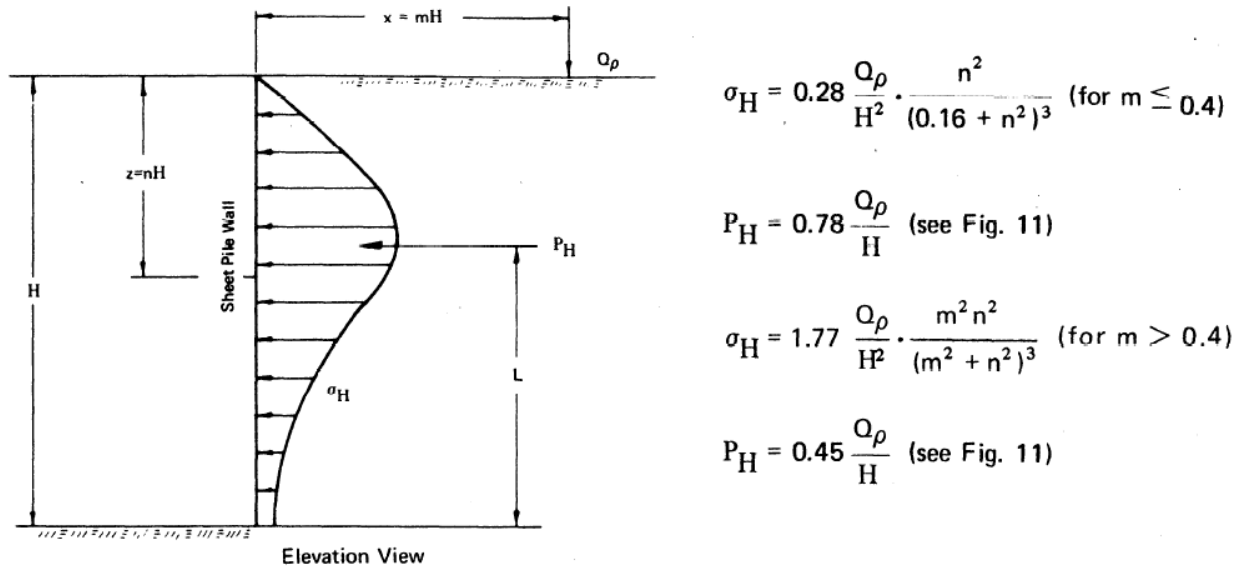


Figure 4. Lateral Pressure Due to Point Load (Elevation) (After Terzaghi)



SHEET PILING

Figure 5. Lateral Pressure Due to Point Load (Plan) (After Terzaghi)

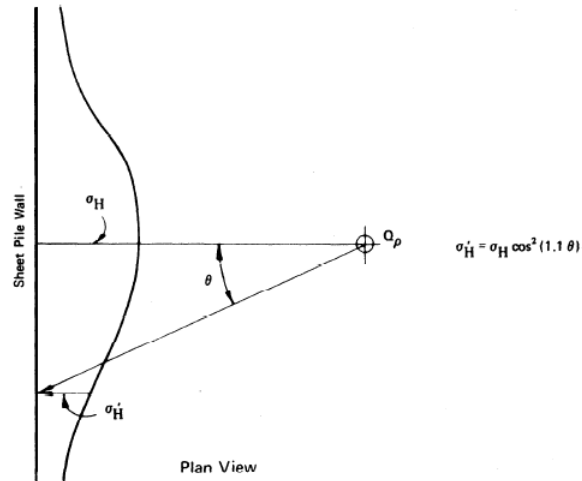
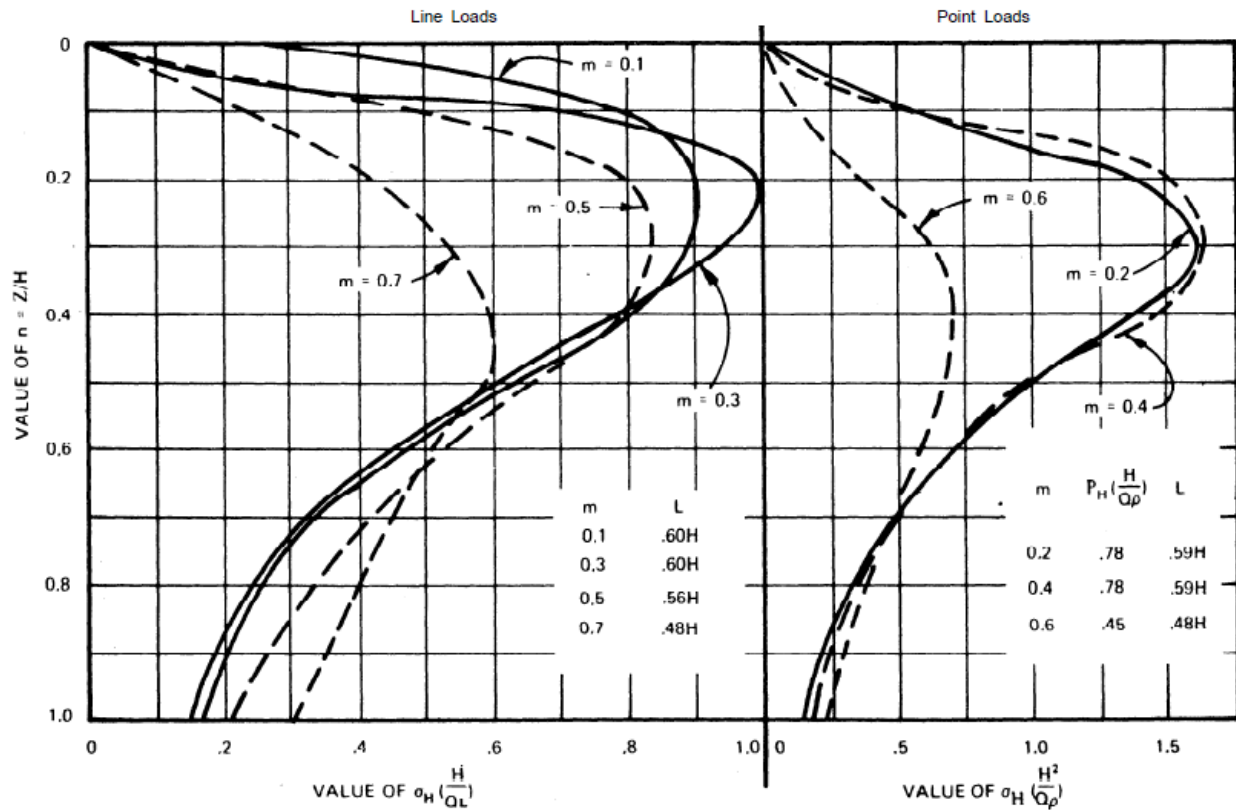
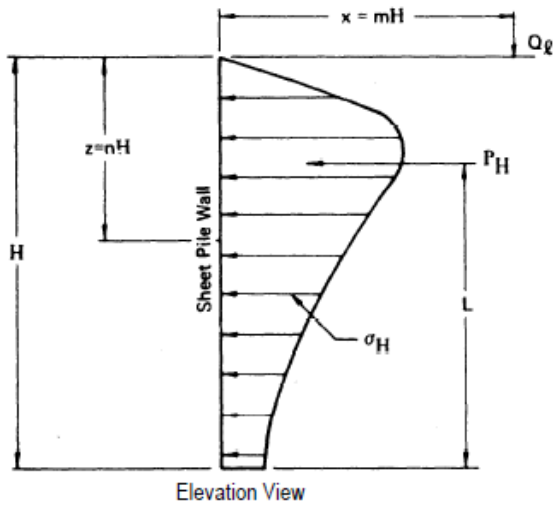


Figure 6. Lateral Pressure Distribution and Location of Resultant Force Due to Point or Line Load (After Navdocks)



SHEET PILING

Figure 7. Lateral Pressure Due to Line Load (After Terzaghi)



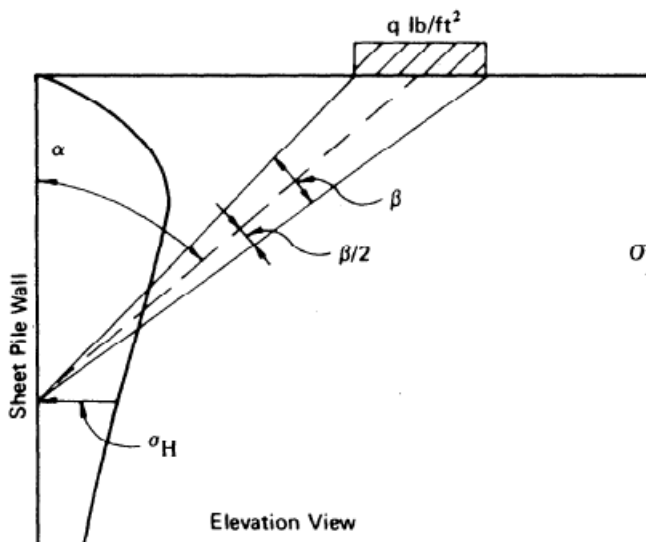
$$\sigma_H = 0.20 \frac{Q_l}{H} \cdot \frac{n}{(0.16 + n^2)^2} \quad (\text{for } m \leq 0.4)$$

$$P_H = 0.55 Q_l, \text{ resultant force}$$

$$\sigma_H = 1.28 \frac{Q_l}{H} \cdot \frac{m^2 n}{(m^2 + n^2)^2} \quad (\text{for } m > 0.4)$$

$$P_H = \frac{0.64 Q_l}{(m^2 + 1)} \text{ resultant force}$$

Figure 8. Lateral Pressure Due to Strip Load (After Teng)



$$\sigma_H = \frac{2q}{\pi} [\beta - \sin \beta \cos 2 \alpha]$$

SHEET PILING

Figure 9. Earth Pressure on Cantilever Sheet Piling (After Teng)

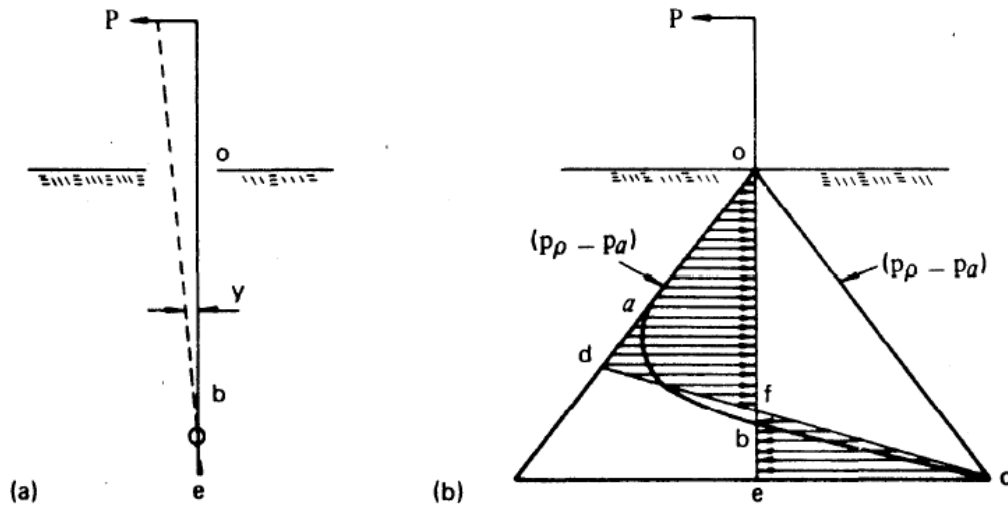
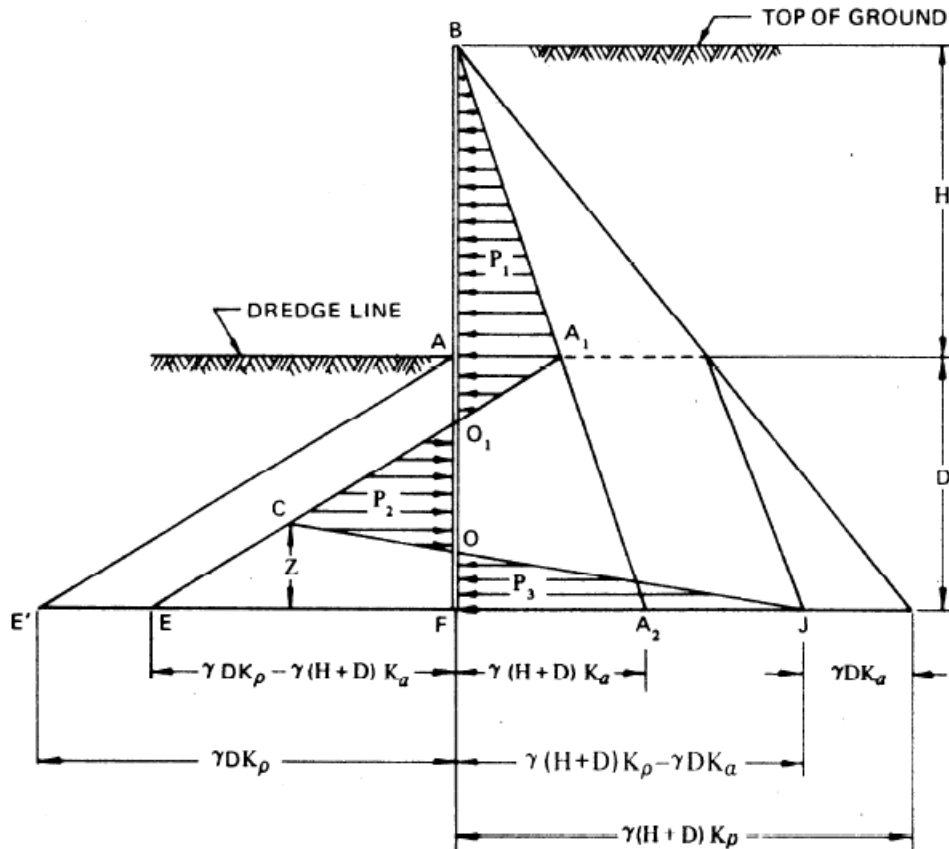


Figure 10. Earth Pressure for Design of Cantilever Sheet Piling in Granular Soil



SHEET PILING

Figure 11. Initial Earth Pressure for Design of Cantilever Sheet Piling Entirely in Cohesive Soil (*After Teng*)

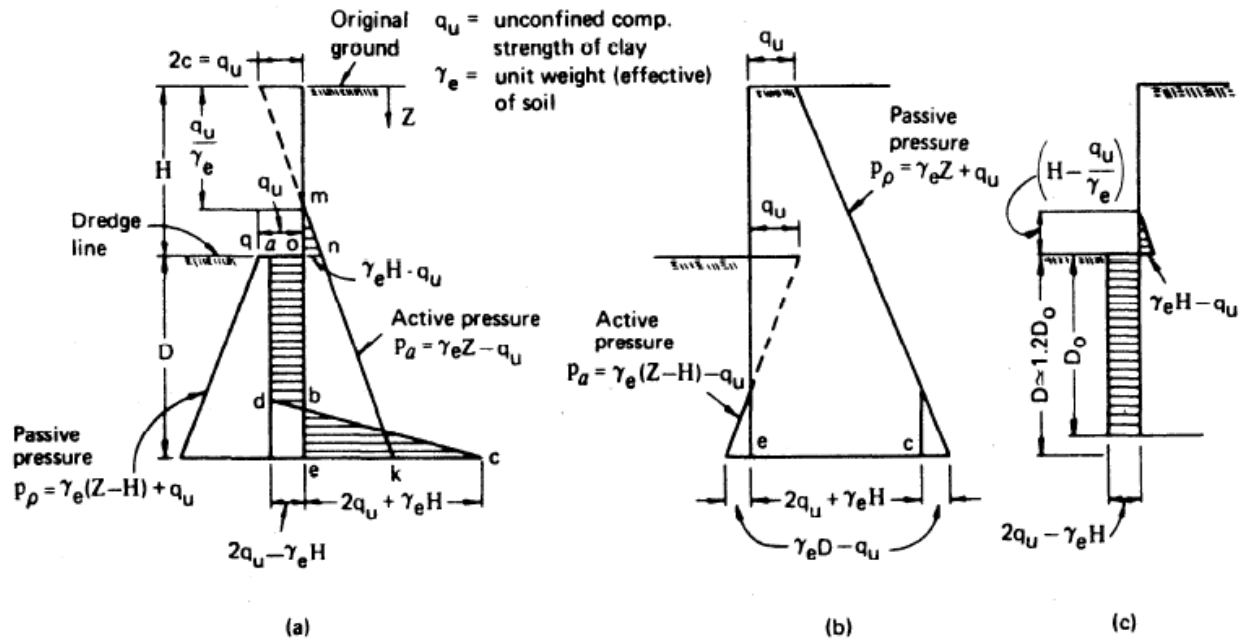
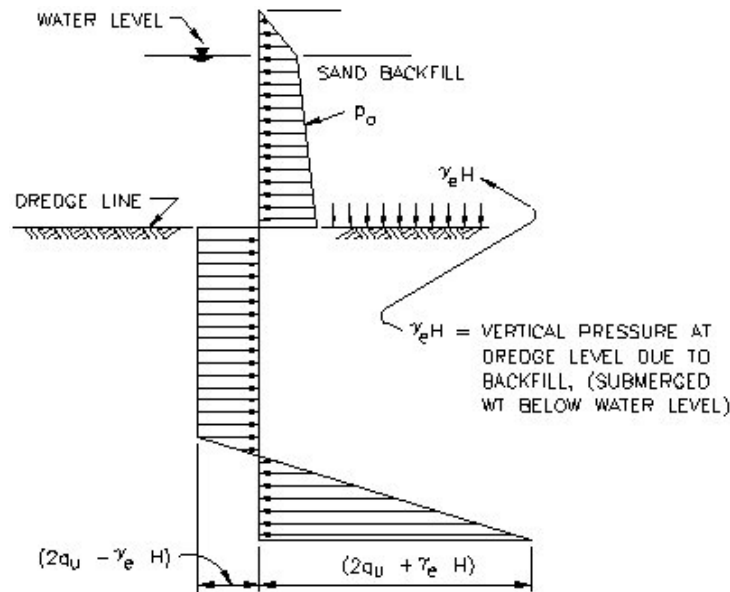


Figure 12. Initial Earth Pressure for Design of Cantilever Sheet Piling in Cohesive Soil with Granular Backfill (*After Teng*)



SHEET PILING

Figure 13. Effect of Depth of Penetration on Pressure Distribution and Deflected Shape

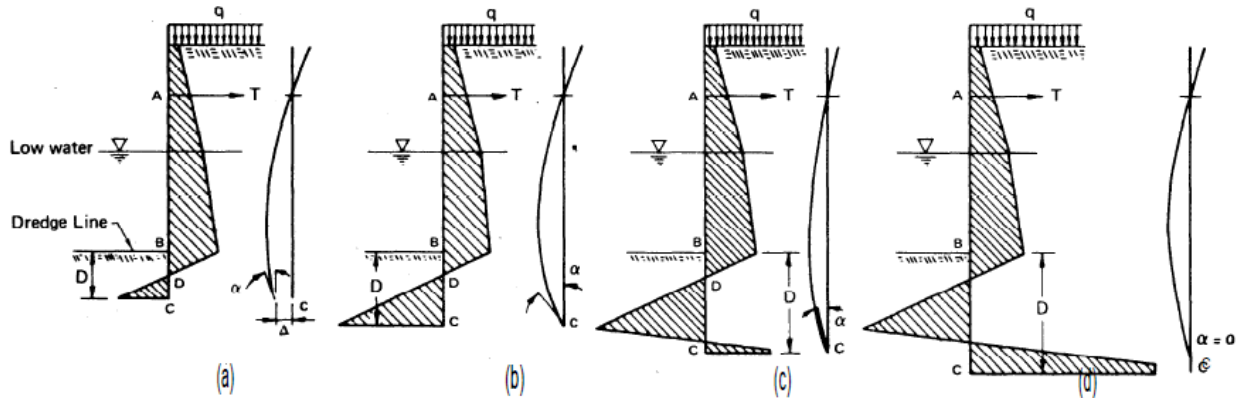


Figure 14. Earth Pressure for Design of Anchored Sheet Piling in Granular Soil by the Free Support Method (After Teng)

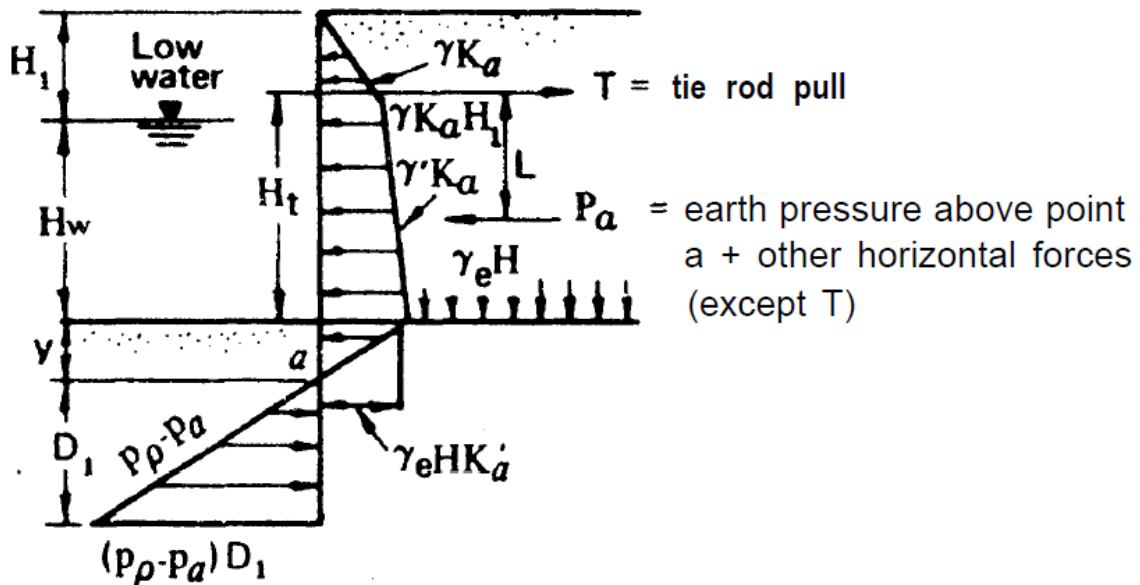
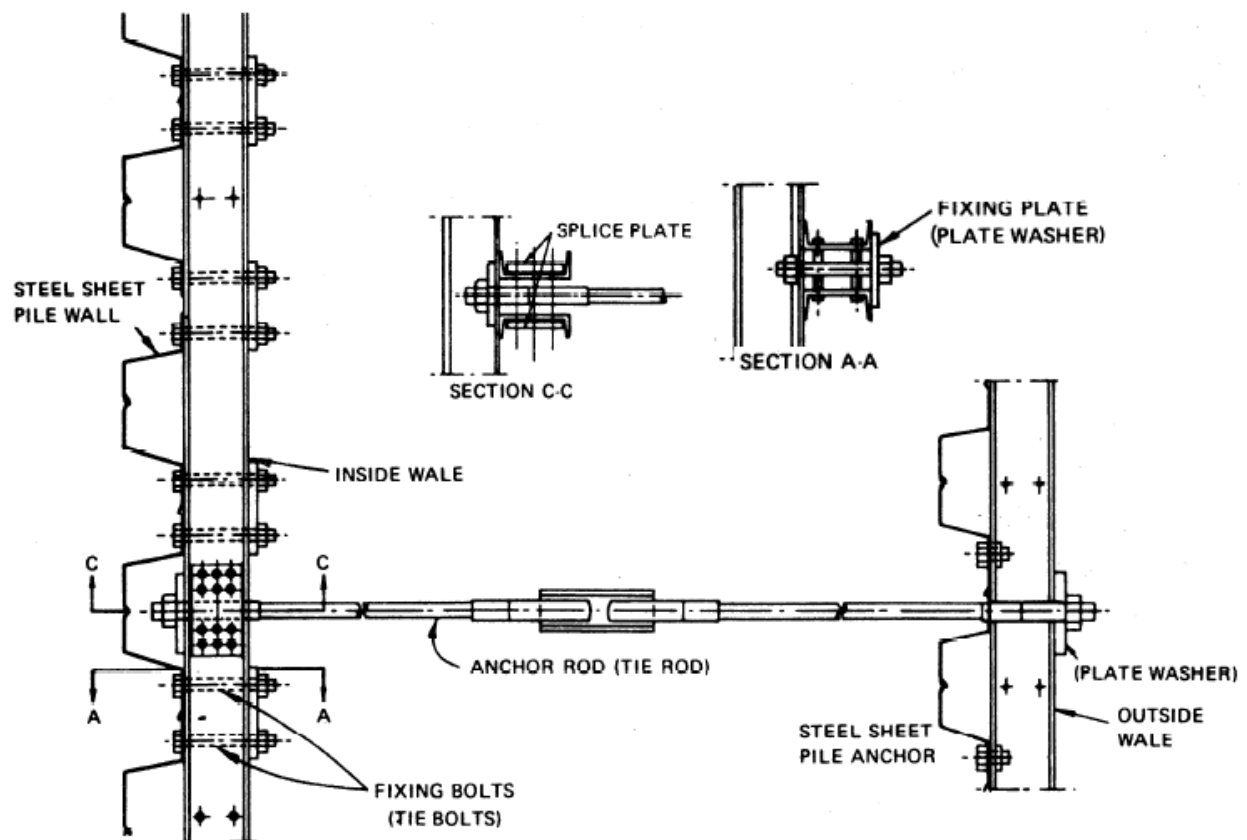


Diagram illustrating the forces and dimensions for a sheetpile wall in a dredge cut. The wall is shown in cross-section, with labels for various components and forces:

- Cantilever or anchored sheetpiling**: The vertical wall structure.
- Dredge line**: The horizontal line indicating the dredge level.
- Backfill or original soil deposit**: The material behind the wall.
- Z**: Vertical distance from the dredge line to the top of the wall.
- H**: Vertical distance from the dredge line to the bottom of the wall.
- q_u** : Uniform surcharge load on the dredge line.
- $\gamma_e H$** : Effective vertical stress at the dredge line.
- $\gamma_e H - q_u$** : Net vertical stress at the dredge line.
- $P_\rho = \gamma_e (Z - H) + q_u$** : Active earth pressure on the backfill side.
- $P_a = \gamma_e H + \gamma (Z - H) - q$** : Active earth pressure on the dredge side.
- $2q_u - \gamma_e H$** : Net horizontal force at the bottom of the wall.

SHEET PILING

Figure 17. Typical Wale and Anchor Rod Details



SHEET PILING

Figure 18. Types of Anchorage Systems (After Teng)

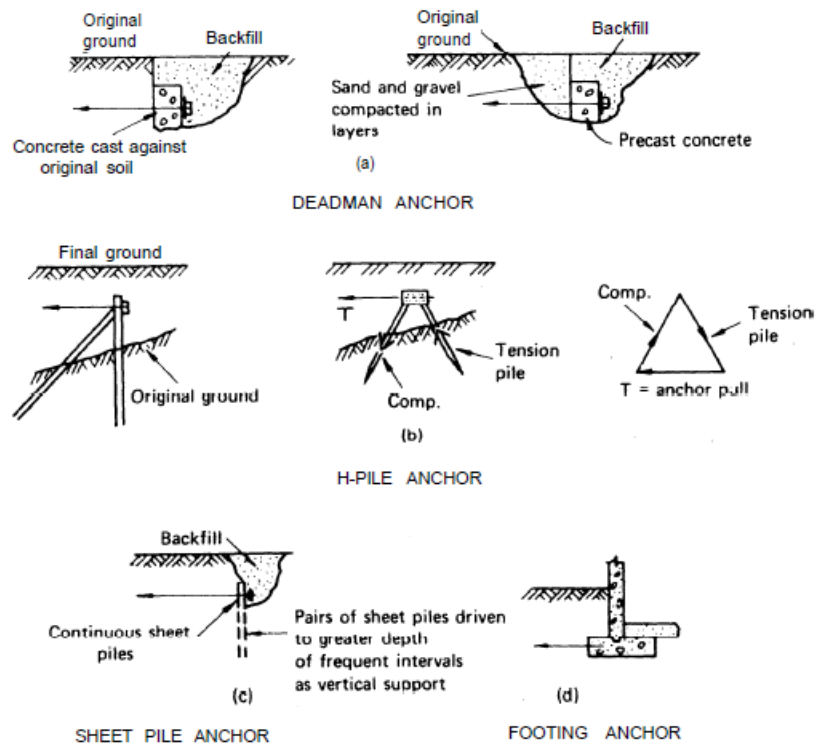
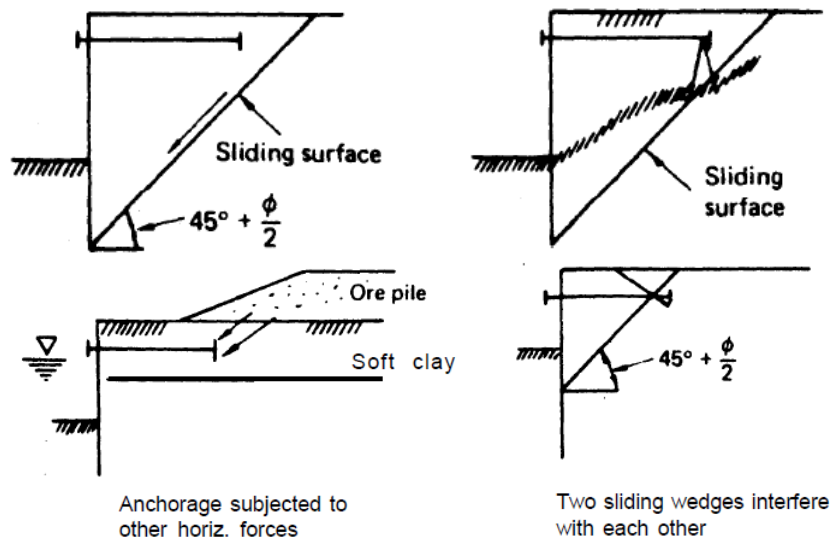


Figure 19. Installations Having Reduced Anchorage Capacity (After Teng)



SHEET PILING

Figure 20. Effects of Anchor Location Relative to the Wall (After Navdocks, Terzaghi)

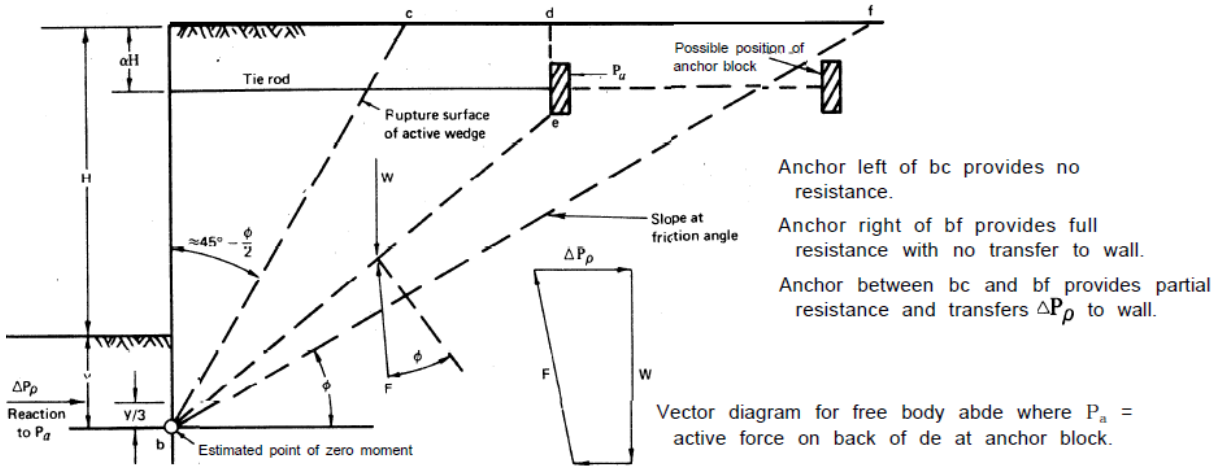
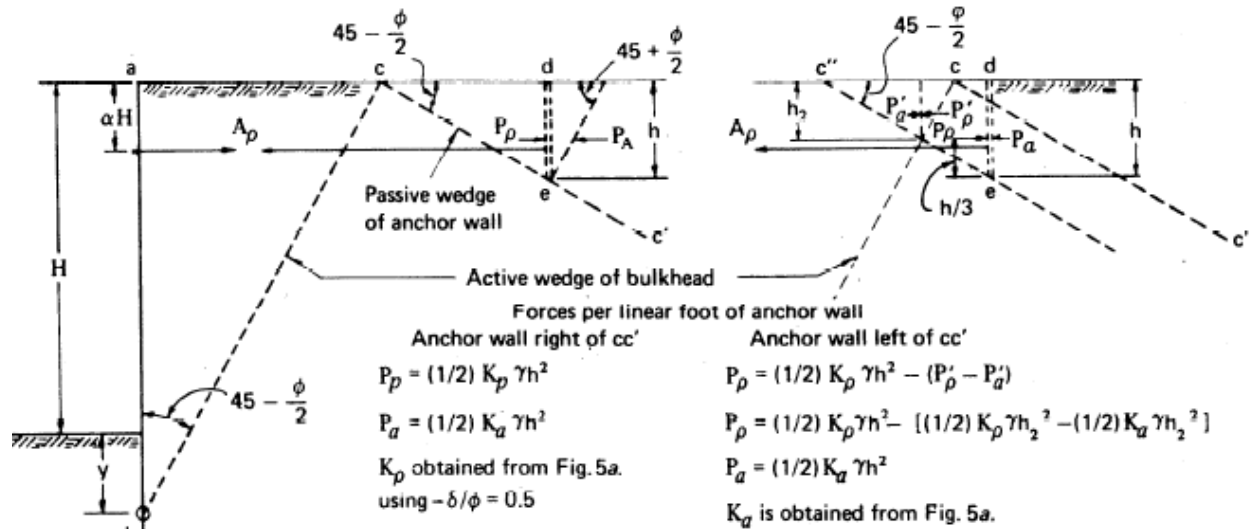


Figure 21. Continuous Anchor Walls (After Navdocks)



SHEET PILING

Figure 22. Continuous Deadmen Near Ground Surface (*After Teng*)

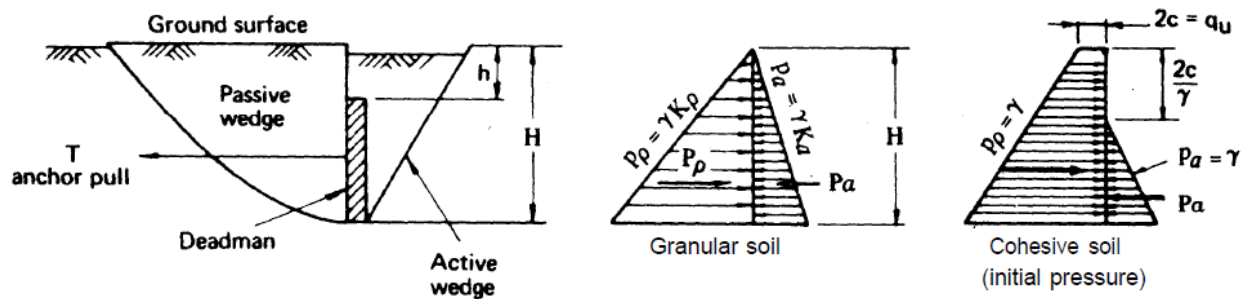
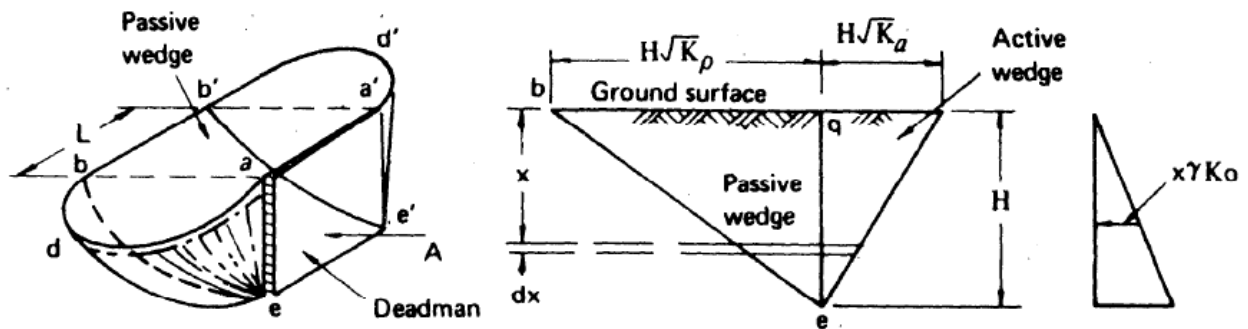


Figure 23. Short Deadmen Near Ground Surface (*After Teng*)



SHEET PILING

Figure 24. Wood Sheet Piles

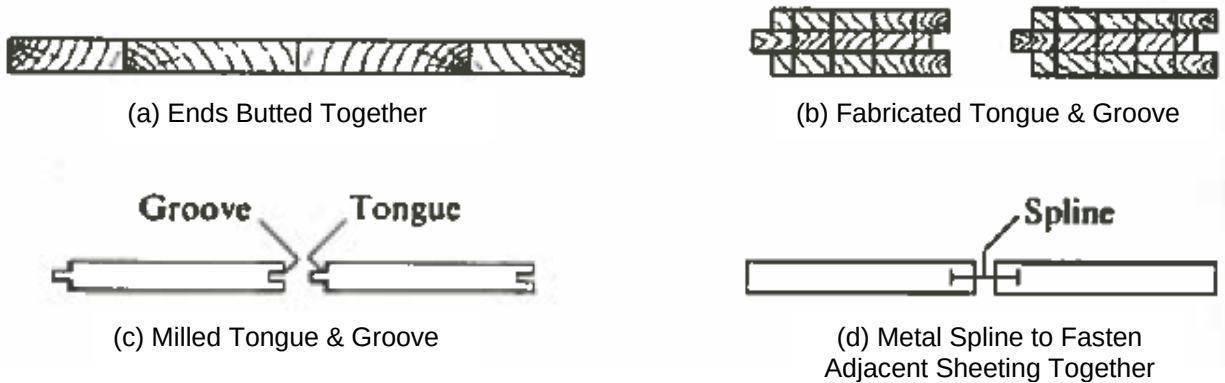
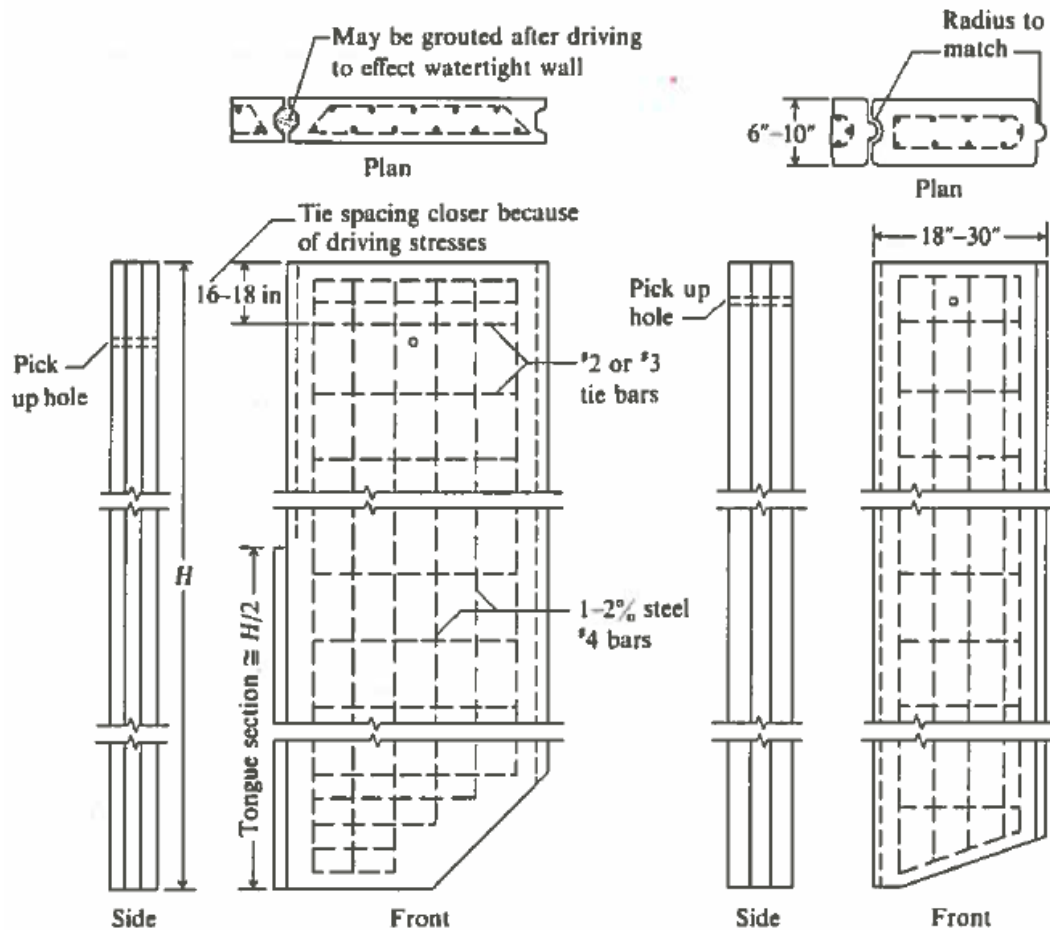
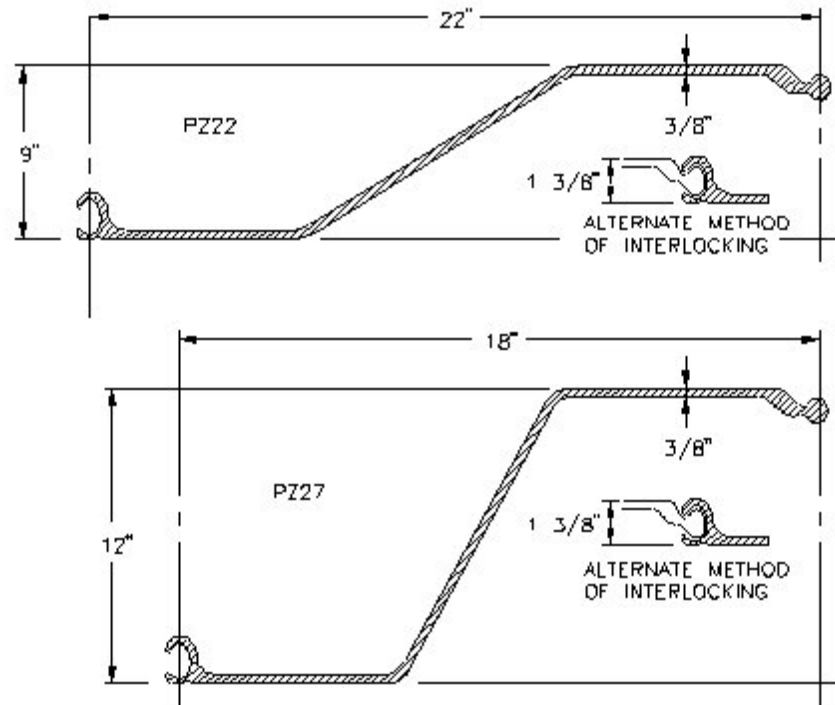


Figure 25. Concrete Sheet Piles



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Figure 26. Steel Sheet Piles



SECTION DESIGNATION	AREA SQ IN	NOMINAL WIDTH IN.	WEIGHT IN POUNDS		MOMENT OF INERTIA IN. ⁴	SECTION MODULUS IN. ³	
			PER LIN FT OF BAR	PER SQ FT OF WALL		SINGLE SECTION	PER LIN FT OF WALL
PZ22	11.86	22	40.3	22	154.7	33.1	18.1
PZ27	11.91	18	40.5	27	276.3	45.3	30.2

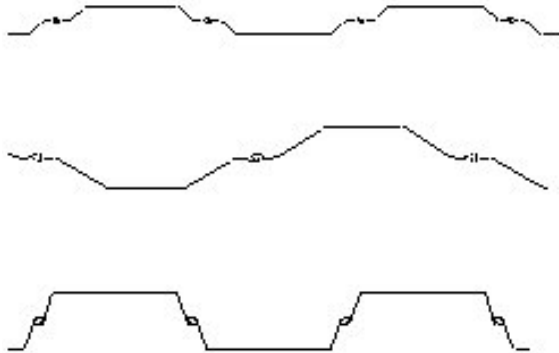
SURFACE AREA SQ FT PER LIN FT OF BAR	
TOTAL AREA	NOMINAL COATING AREA
4.94	4.48
4.94	4.48

MECHANICAL PROPERTIES			
GRADE	MIN YIELD POINT Ksi	MIN TENSILE STRENGTH Ksi	ELONGATION IN. 8 IN. %
A328	39	70	17
A572-Gr50	50	65	18
A690	50	70	18

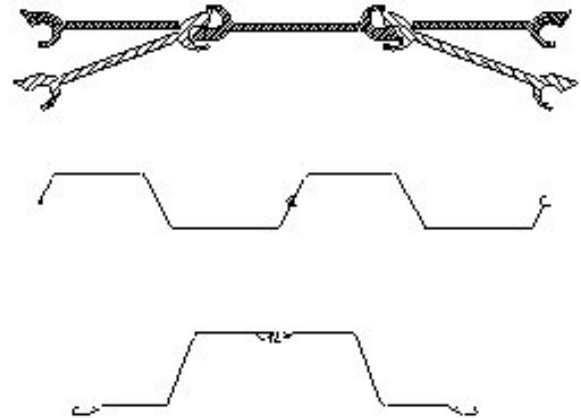
(a) Some Bethlehem Steel Piles



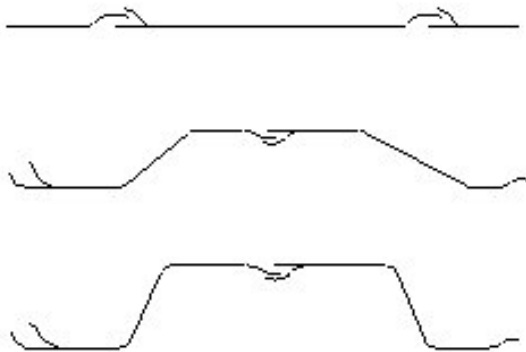
SHEET PILING



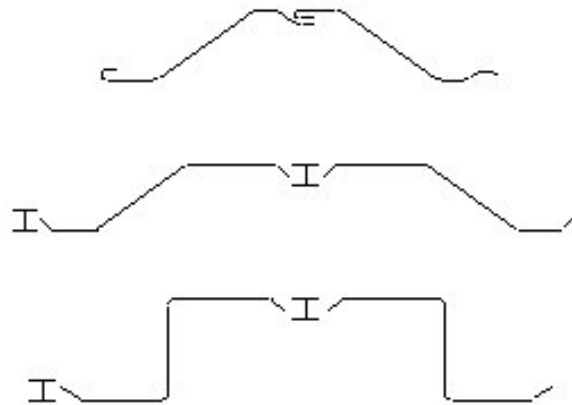
(a) Larson Piles
(After Western Service Corp.)



(b) Arbed-Belval Piles
(After Amerlux Steel Products Corp.)



(c) Hoesch Piles



(d) Krupp Piles

(b) Some European Steel Sheet Piles

SHEET PILING

Figure 27. Typical Fabricated or Rolled Steel Sheet Pile Joints

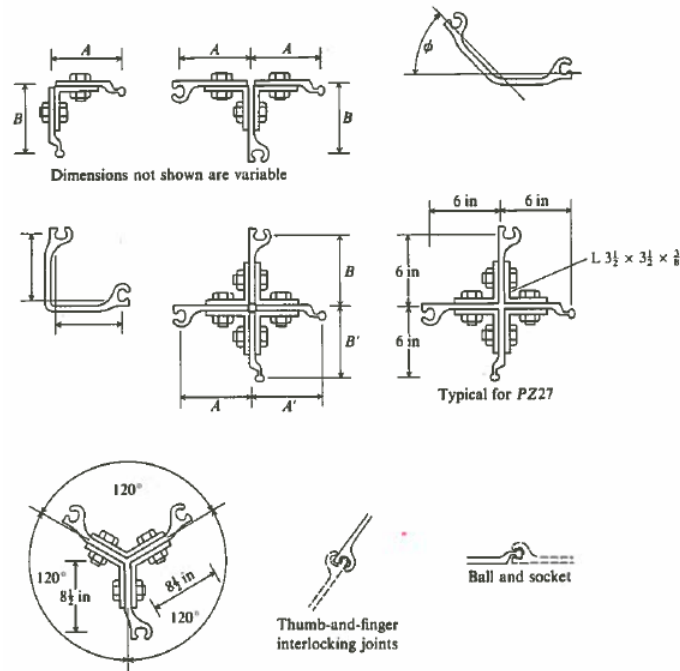
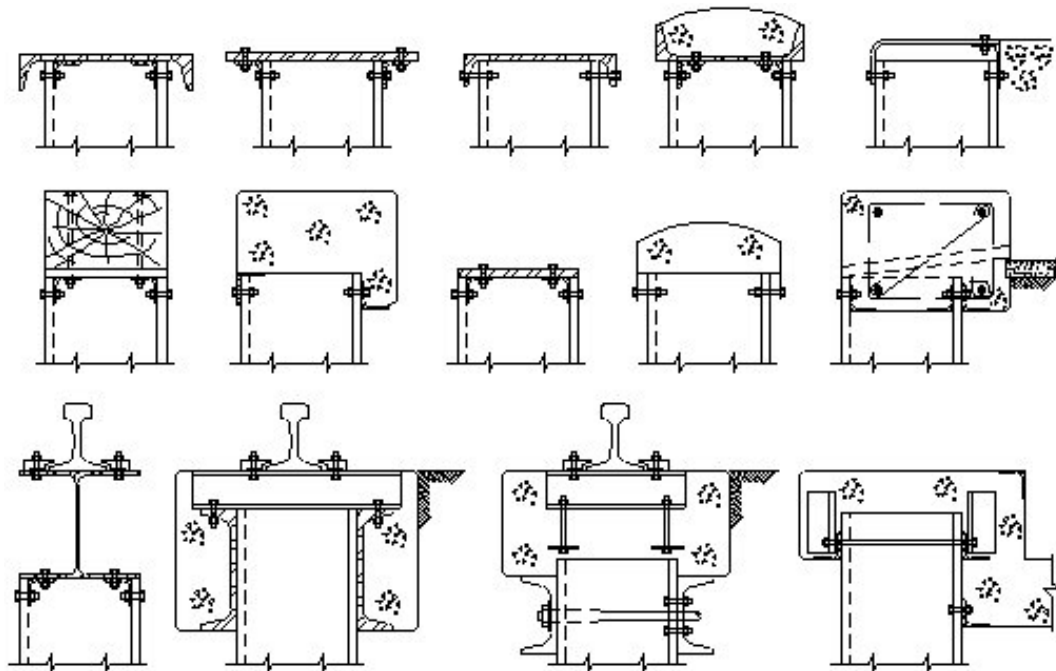


Figure 28. Typical Caps for Steel Sheet Piling





SHEET PILING

Sample Design 1: Cantilevered Sheet Pile Wall: Granular Soil

The following example originated from the USS Steel Sheet Piling Design Manual.

(Some coefficients and calculations have been revised to reflect the Rankine Theory.)

Medium Sand

$$\gamma = 115 \text{ pcf}$$

$$\gamma' = 65 \text{ pcf}$$

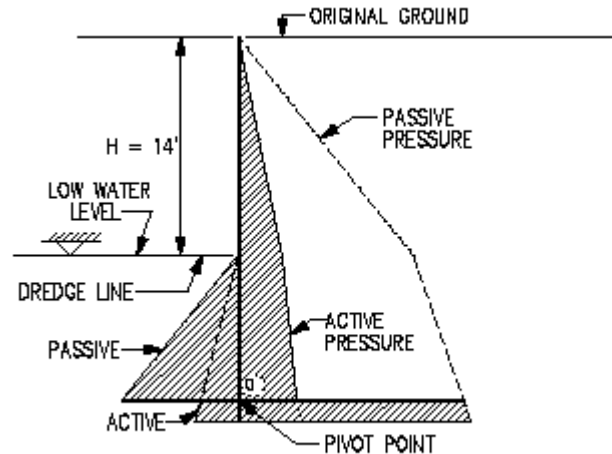
$$\phi = 35^\circ$$

$$K_a = 0.27$$

$$K_p = 3.69$$

Apply safety factor at end:

$$K_p - K_a = 3.42$$



Determine Wall Pressures

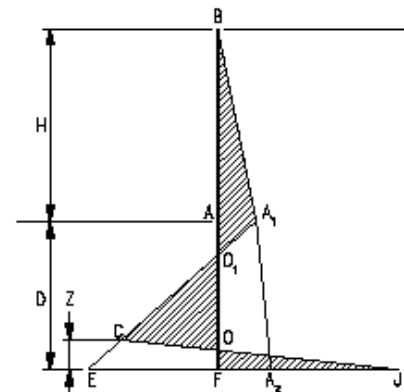
$$P_{A1} = \gamma H K_a = (115)(14.0)(0.27) = 435 \text{ psf}$$

$$\begin{aligned} P_{A2} &= P_{A1} + \gamma' D K_a = 435 + (65)(0.27)D \\ &= 435 + 17.6D \end{aligned}$$

$$\begin{aligned} P_E &= \gamma' D (K_p - K_a) - P_{A1} = 65D(3.42) - 435 \\ &= 222D - 435 \end{aligned}$$

$$\begin{aligned} P_J &= \gamma' D (K_p - K_a) + \gamma H K_p \\ &= 65D(3.42) + 115(14)(3.69) \\ &= 222D + 5941 \end{aligned}$$

CONVENTIONAL ASSUMED PRESSURE DIAGRAM



From Statics, the following conditions must be satisfied

(1) $\sum F_H = 0$ in terms of areas:

$$\text{Area}(B A A_1) + \text{Area}(A A_1 A_2 F) + \text{Area}(E C J) - \text{Area}(E A_1 A_2) = 0$$

OR

$$\frac{1}{2}(H)P_{A1} + (P_{A1} + P_{A2})\frac{D}{2} + (P_E + P_J)\frac{Z}{2} - (P_E + P_{A2})\frac{D}{2} = 0$$



SHEET PILING

Sample Design 1: Cantilevered Sheet Pile Wall: Granular Soil

Solving for Z:
$$Z = \frac{(P_E + P_{A1})D - HP_{A1}}{P_E + P_J}$$

(2) ΣM about any point is zero

$$\Sigma M_F = \frac{1}{2}(H)(P_{A1})(D + \frac{H}{3}) + (P_{A1})\frac{D^2}{2} + (P_E + P_J)\frac{Z^2}{6} - (P_E + P_{A2})\frac{D^2}{6} + (P_{A2} - P_{A1})\frac{D^2}{6} = 0$$

Method of Solution:

1. Assume a Depth of Penetration, D
2. Calculate Z
3. Substitute Z into ΣM , and check if zero. Adjust D and recalculate if necessary.

Try D = 14.75 ft.

$$P_{A1} = 435 \text{ psf} \quad P_{A2} = 695 \text{ psf} \quad P_J = 9216 \text{ psf} \quad P_E = 2840 \text{ psf}$$

$$Z = \frac{(2840 - 435)(14.75) - (14)(435)}{2480 + 9216} = \frac{29,384}{12,065} = 2.44 \text{ ft}$$

$$\begin{aligned} \Sigma M_F &= \frac{1}{2}(14)(435)(14.75 + \frac{14}{3}) + (435)\frac{(14.75)^2}{2} + (695 - 435)\frac{(14.75)^2}{6} \\ &+ (2840 + 9216)\frac{(2.44)^2}{6} - (2840 + 695)\frac{(14.75)^2}{6} \end{aligned}$$

$$= 59,124 + 47,320 + 9428 + 11,963 - 128,181$$

$$= -346 \text{ ft.-lb}$$

SAY O.K.

USE D = 14.75ft.

To assure a margin of safety, D may be increased by 20 to 40% or, alternately, a reduced passive earth pressure coefficient could be used.

USE D = 19.0 ft. (INCREASE = 28.8%)



SHEET PILING

Sample Design 1: Cantilevered Sheet Pile Wall: Granular Soil

MAXIMUM MOMENT AND SHEET PILE SIZE

Locate Point of Zero Shear:

$$Y = \frac{P_{A1}}{\gamma' (K_p - K_a)} = \frac{435}{65(3.42)} = 1.96 \text{ ft.}$$

SAY 2.0 ft.

$$P_1 = \frac{1}{2} P_{A1} H = \frac{1}{2} (435)(14) = 3045 \text{ lb}$$

$$P_2 = \frac{1}{2} P_{A1} Y = \frac{1}{2} (435)(2.0) = 435 \text{ lb}$$

$$\frac{1}{2} \gamma' (K_p - K_a) X^2 = P_1 + P_2$$

$$X^2 = \frac{2(P_1 + P_2)}{\gamma' (K_p - K_a)} = \frac{2(3045 + 435)}{65(3.42)} = \frac{2(3480)}{222}$$

$$X = 5.6 \text{ ft.}$$

Maximum Moment:

$$P_3 = \frac{1}{2} \gamma' (K_p - K_a) (5.6)^2 = P_1 + P_2 = 3480 \text{ lb}$$

$$M_{MAX} = P_1 l_1 + P_2 l_2 - P_3 l_3 \quad l_1 = \left(\frac{H}{3} + Y + X \right)$$

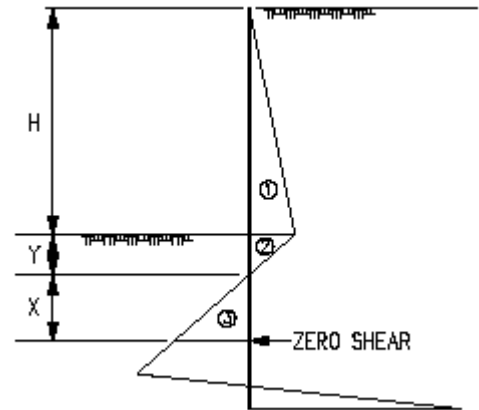
$$l_2 = \left(\frac{24}{3} + X \right)$$

$$l_3 = \frac{X}{3}$$

$$M_{MAX} = 3045 \left(\frac{14}{3} + 2 + 5.6 \right) + 435 \left(\frac{2(2)}{3} + 5.6 \right) - 3480 \left(\frac{5.6}{3} \right) = 37,350 + 3015 - 6495 \quad M_{MAX} = 33,870 \text{ ft.-lbs}$$

Try regular carbon grade; $f_s = 25 \text{ ksi}$

$$\text{Required Section Modulus} = \frac{M}{f_s} = \frac{33,870 \times 12}{25000} = 16.3 \text{ in}^3$$



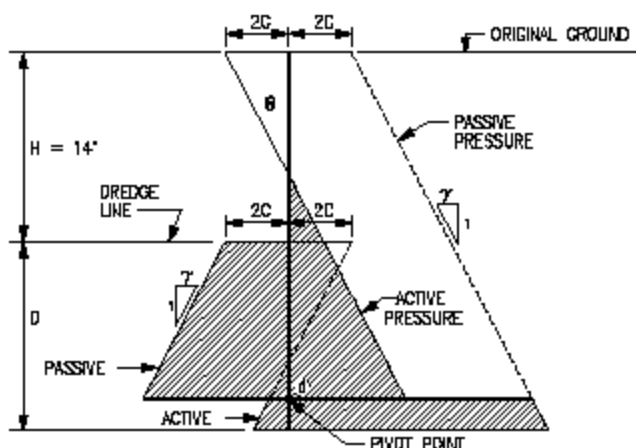


SHEET PILING

Sample Design 2: Cantilevered Sheet Pile Wall: Cohesive Soil

The following example originated from the USS Steel Sheet Piling Design Manual. Only the portion related to the determination of lateral pressures are presented here.

RESULTANT PRESSURE DISTRIBUTION



Medium Soft Clay

$$\gamma = 120 \text{ pcf}$$

$$\gamma' = 60 \text{ pcf}$$

$$(C = 750 \text{ psf})$$

$$\text{Use } C = 500 \text{ psf}$$

$$q_u = 2C = 1000 \text{ psf}$$

$$\phi = 0^\circ$$

Final Strength (Long Term)

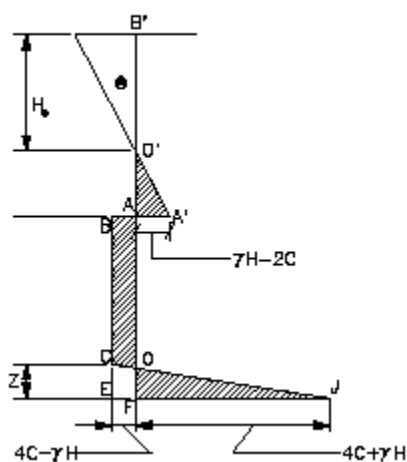
$$C = 0$$

$$\phi = 27^\circ$$

Check Critical Height:

$$H_c = \frac{4C}{\gamma} = \frac{4(500)}{120} = 16.7 \text{ ft} > H$$

CONVENTIONAL ASSUMED PRESSURE DIAGRAM



Determine Wall Pressures

$$H_o = \text{Point of Zero Pressure} = \frac{2C}{\gamma} = \frac{1000 \text{ psf}}{120 \text{ pcf}} = 8.3 \text{ ft}$$

$$H - H_o = 14.0 - 8.3 = 5.7 \text{ ft.}$$

$$\gamma H - 2C = 120(14.0) - 1000 = 680 \text{ psf}$$

$$4C - \gamma H = 4(500) - 120(14.0) = 320 \text{ psf}$$

$$4C + \gamma H = 4(500) + 120(14.0) = 3680 \text{ psf}$$

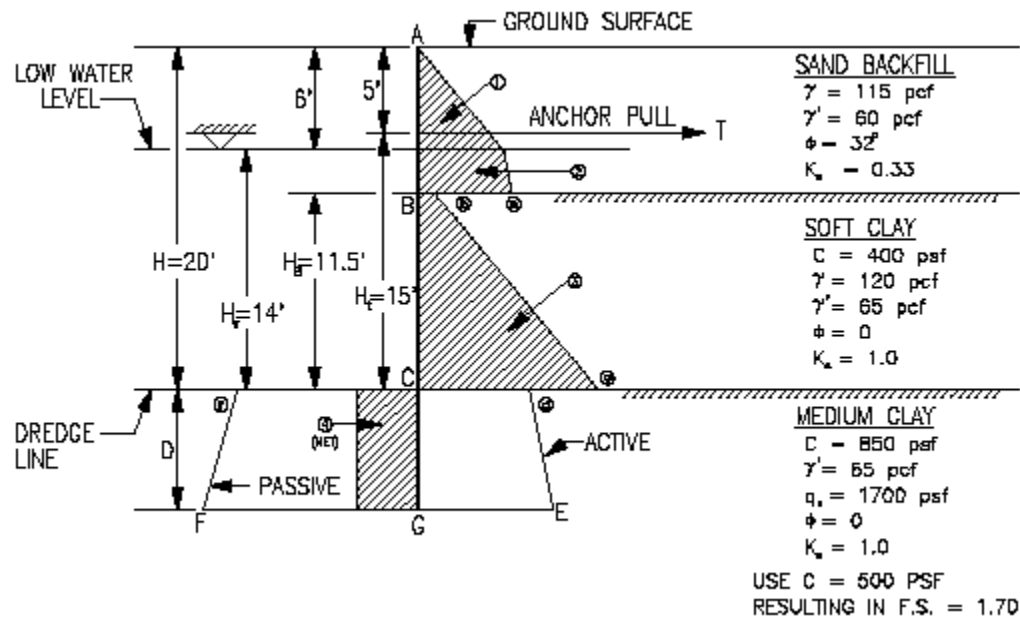


SHEET PILING

Sample Design 3: Design of Anchor Sheet Pile Wall - Cohesive Soil with Sand Backfill - Free Earth Support Method

The following example originated from the USS Steel Sheet Piling Manual.

RESULTANT PRESSURE DISTRIBUTION



Determine Wall Pressures

$$P_{Ba} = \gamma(H - H_w)K_a + \gamma'(H_w - H_B)K_a = 115(6)(0.33) + 60(2.5)(0.33) = 228 + 50 = 278 \text{ psf}$$

$$P_{Bb} = \gamma(H - H_w) + \gamma'(H_w - H_B) - 2C = 115(6) + 60(2.5) - 2(400) = 40 \text{ psf}$$

$$P_{Cc} = P_{Bb} + \gamma'(H_B) = 40 + 65(11.5) = 788 \text{ psf}$$

$$P_{Cd} = \Sigma\gamma(H) - 2C = 115(6) + 60(2.5) + 65(11.5) - 2(500) = 588 \text{ psf}$$

$$P_{Cf} = 2C = 2(500) = 1000 \text{ psf}$$

$$P_{Cnet} = P_{Cf} - P_{Cd} = 1000 - 588 = 412 \text{ psf}$$

Resultant of Pressure Distribution (see numbered areas)

$$P_1 = \frac{115(6)^2(0.33)}{2} = 684 \text{ lb.}$$

$$P_2 = \frac{(228 + 278)}{2} (2.5) = 633 \text{ lb.}$$

$$P_3 = \frac{(40 + 788)}{2} (11.5) = 4760 \text{ lb.} \quad \text{C.G. @ } 8.5 + 7.48 = 15.98 \text{ ft. from A}$$

$$P_4 = 412D$$



SHEET PILING

Sample Design 3: Design of Anchor Sheet Pile Wall - Cohesive Soil with Sand Backfill - Free Earth Support Method

From Statics, the following conditions must be satisfied:

$$(1) \sum F_H = P_1 + P_2 + P_3 - P_4 - T = 0$$

(2) $\sum M$ about the anchor tie rod must be zero:

$$\sum M_{AP} = 684(1) - 633(1 + 1.29) - 4760(15.98 - 5.0) + 412D \left(\frac{D}{2} + 15 \right) = 0$$

OR

$$D^2 + 30D = 257.4$$

Solving $D = 7.0$ ft.

MAKE $D = 10$ FT. FOR ADDED SAFETY AGAINST
OVERDREDGING

Tensile force in tie rod is given by:

$$T = 6077 - 412(7) = 3193 \text{ lb.} \quad \text{SAY } 3200 \text{ lb/ft (without F.S.)}$$

A bulkhead of this type (in clay) should be checked for longer-term soil characteristics and for possible deep seated rotational failure.

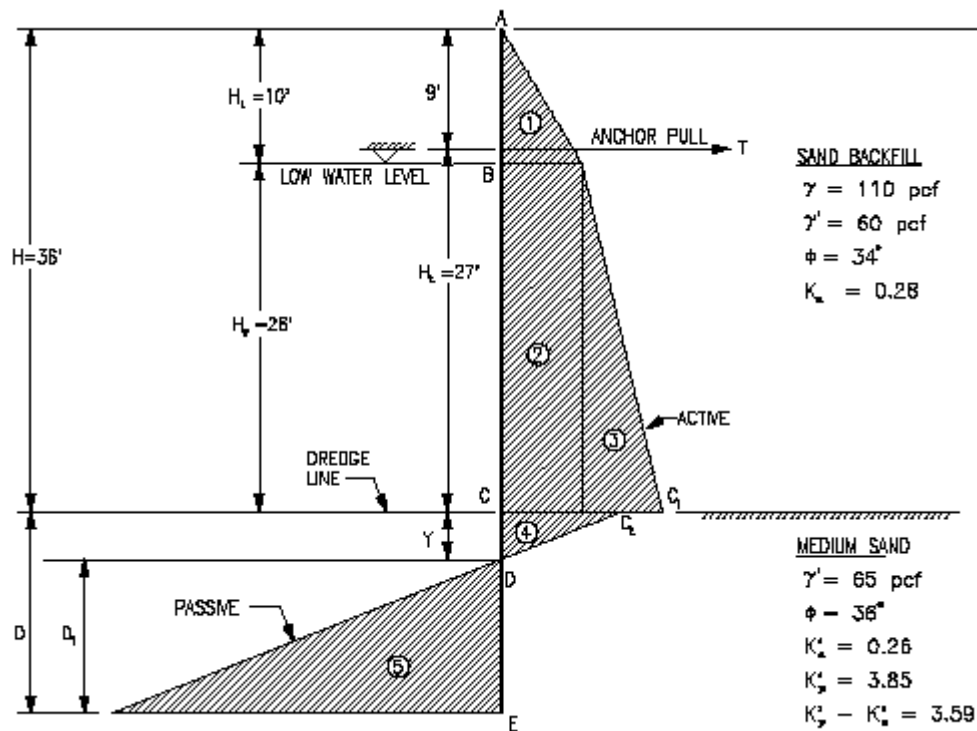


SHEET PILING

Sample Design 4: Anchored Sheet Pile Wall Granular Soil - Free Earth Support Method

The following example originated from the USS Steel Sheet Piling Design Manual.

(Some coefficients and calculations have been revised to reflect the Rankine Theory)



Determine Pressures on Wall

$$P_B = \gamma H_1 K_a = (110)(10)(0.28) = 308 \text{ psf}$$

$$P_{C1} = P_B + \gamma' H_w K_a = 308 + 60(26)(0.28) = 308 + 437 = 745 \text{ psf}$$

$$P_{C2} = [\gamma H_1 + \gamma' H_w] K'_a = [110(10) + 60(26)](0.26) = 692 \text{ psf}$$

$$P_E = \gamma' (K'_p - K'_a) D_1 = 65(3.59)D_1 = 233D_1$$

Calculate point of zero pressure, y

$$y = 692 / (3.59 \times 65) = 2.97 \text{ ft}$$

Resultant Pressure Distribution (see numbered areas)

$$P1 = 308 \times 10 / 2 = 1540 \text{ lbs}$$

$$P2 = 308 \times 26 = 8008 \text{ lbs}$$



SHEET PILING

Sample Design 4: Anchored Sheet Pile Wall Granular Soil - Free Earth Support Method

$$P3=437 \times 26/2=5681 \text{ lbs}$$

$$P4=692 \times 2.97/2=1028 \text{ lbs}$$

$$P5=233(D_1)^2/2=116.5(D_1)^2$$

Take moments about tie rod level:

$$-1540(10/3-1)+8008(26/2+1)+5681(26 \times 2/3+1)+1028(2.97/3+27)-116.5(D_1)^2(27+2.97+D_1 \times 2/3)=0$$

which reduces to:

$$241444.1-3491.5(D_1)^2-77.7(D_1)^3=0$$

Solve D_1 by trial:

$$D_1=7.685 \text{ ft}$$

add 20 to 40 percent to D_1 to provide safety margin

$$\text{use } D_1=7.685 \times 1.3=10 \text{ ft.}$$

Calculate tie rod tension:

$$1540+8008+5681+1028-116.5(7.685)^2-T=0$$

$$T=9377 \text{ lbs/ft} \times 1.3=12200 \text{ lbs/ft} \text{ use for design of tie rod and wale}$$